# The second Bougard–Joret line to order 24: an exhaustive check, the extremal graphs at 35 triples, and a construction without a clique partition for every odd $\alpha$

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github.com/fsantibanezleal/CAOS_RESEARCH

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**Abstract**

Let $f(n, \alpha, k)$ be the least number of edges of a $k$-connected graph with $n$ vertices and independence number $\alpha$. Bougard and Joret conjectured $f(n, \alpha, k) = t(n, \alpha) + \lceil k\alpha/2 \rceil$ for $n > k\alpha$, where $t(n, \alpha)$ counts the edges of $\alpha$ disjoint balanced cliques, and Didin has announced a proof of the lower bound. We give an independent exhaustive check. At every one of the 53 triples with $\alpha \geq 3$, $k \geq 3$ and $n \leq 24$ outside the range that Bougard and Joret's Proposition 8 settles, $f$ equals the conjectured value, by complete enumeration of 6,223,452,780 candidate graphs with nauty's generator, an independence prune proved sound, and an exact filter. For seven pairs $(\alpha, k)$, including $(5, 3)$, $(4, 4)$ and $(3, 6)$, this settles the conjectured value for every $n > k\alpha$. A validation suite checks every component: rebuilt binaries are byte-identical to the production ones, 26,005 filter decisions on graphs up to order 24 and 27,132 prune decisions agree with an independent implementation, and the generator's splitting and counts agree with known values. We list all 246 extremal graphs at 35 triples. Those that split into $\alpha$ cliques have Bougard and Joret's structure; the other 40 do not, and all have $k = 3$ and $\alpha$ odd. Conditional on Didin's main theorem, we prove that such graphs occur only at $k = 3$ with $\alpha$ odd and $n \leq 4\alpha - 1$, or at six small triples, as the data show. Two blocks recovered from the data give, for every odd $\alpha$ and $3\alpha < n \leq 4\alpha - 1$, an explicit 3-connected graph with the conjectured number of edges, independence number $\alpha$ and no partition into $\alpha$ cliques; it is extremal at order 24 and below, and everywhere if Didin's theorem holds.

## 1 Introduction

All graphs are finite and simple. For a graph $G$ write $\mathrm{e}(G)$ for its number of edges, $\alpha(G)$ for its independence number, $\kappa(G)$ for its vertex connectivity and $\theta(G)$ for its clique cover number, the least number of cliques whose vertex sets partition $V(G)$. For integers $n, \alpha, k$ let $f(n, \alpha, k)$ be the

least number of edges of a $k$-connected graph with $n$ vertices and independence number exactly $\alpha$, and let

$$t(n,\alpha) = r\binom{q+1}{2} + (\alpha - r)\binom{q}{2}, \qquad n = q\alpha + r,\ 0 \le r < \alpha, \tag{1}$$

be the number of edges of the Turán-type graph $T(n,\alpha)$, the disjoint union of $\alpha$ cliques whose orders differ by at most one. By Turán's theorem [29] in its complementary form, every graph with $n$ vertices and independence number at most $\alpha$ has at least $t(n,\alpha)$ edges, with equality only for $T(n,\alpha)$.

Ore asked for the least size of a connected graph with given order and independence number. Christophe et al. [8] and Gitler and Valencia [15] settled it independently; Bougard and Joret gave a short proof with a description of all extremal graphs, and answered the 2-connected question in the same way [2, Propositions 5 and 6]. For $k \ge 3$ they conjectured [2, Conjecture 1] that, for $n \ge 2\alpha$, $n \ge \alpha + k$, $\alpha \ge 2$ and $k \ge 3$,

$$f(n,\alpha,k) = \begin{cases} \lceil nk/2 \rceil, & n \le k\alpha, \\ t(n,\alpha) + \lceil k\alpha/2 \rceil, & n > k\alpha. \end{cases} \tag{2}$$

The first line is the degree lower bound $\lceil nk/2 \rceil$. The second is an upper bound by construction: $T(n,\alpha)$ becomes $k$-connected after adding $\lceil k\alpha/2 \rceil$ edges between its cliques without lowering the independence number (Section 2). Das and Gupta determined $f$ on the boundary $n = \alpha + k$ and found that the first line fails there, the smallest failure being $f(7,3,4) = 15 > 14$ [9]. The second line is the subject of this paper; we call it the *second line* and the triples with $n > k\alpha$ the *second regime*.

Bougard and Joret proved the second line when $n$ is large compared with $k\alpha^2$: their Proposition 8 forces a partition into $\alpha$ cliques below a size threshold, and $k$-connectivity then forces $\lceil k\alpha/2 \rceil$ edges between the cliques [2, p. 12]. The triples of the second regime outside that range form what we call the *open window*. Didin has since announced a proof of the lower bound $f(n,\alpha,k) \ge t(n,\alpha) + \lceil k\alpha/2 \rceil$ for every $n > k\alpha$ and $k \ge 3$ [10, Theorem 9], as a consequence of a general Turán-type inequality for graphs with a prescribed independence number [10, Theorem 3]; the preprint is under review, and a short announcement appears in [11]. With Bougard and Joret's construction, a correct proof settles the second line.

**Contributions.** This paper gives an independent, exhaustive and validated computational check of the second line on every open-window triple with $n \le 24$, and determines the extremal graphs.

1. *Values.* For every one of the 53 open-window triples with $\alpha \ge 3$, $k \ge 3$ and $n \le 24$ we decide $f(n,\alpha,k)$ by enumerating every candidate graph with fewer edges than the conjectured value (Theorem 8). The search examines 6,223,452,780 graphs in 42.4 hours of wall-clock time, summed over the edge counts. For 7 pairs $(\alpha,k)$ the window ends by order 24, so the second line holds there for every $n > k\alpha$ (Corollary 9).

2. *Method and soundness.* The search is a finite decision procedure (Proposition 2) built from nauty's generator `geng` [23, 22], a pruning hook that discards partial graphs with too large an independent set (Lemma 3), and an exact filter (Algorithms 1 and 2).

3. *Validation.* Every component is tested against an independent implementation, known counts or a byte-for-byte rebuild of the binaries, on slices of the production streams up to order 23 and on constructed graphs up to order 24 (Section 5).

4. *Extremal graphs.* At 35 triples we list every extremal graph. A short lemma shows that an extremal graph that splits into $\alpha$ cliques has exactly Bougard and Joret's structure (Lemma 10); 40 of the 246 extremal graphs do not split, and have $\theta(G) = \alpha + 1$.

5. *Where the other extremal graphs live.* Conditional on Didin's main theorem, an extremal graph without a partition into $\alpha$ cliques exists only for $k = 3$, $\alpha$ odd and $n \leq 4\alpha - 1$, or at six triples with $n = k\alpha + 1$ or $n = k\alpha + 2$ (Proposition 12). Every observed one lies in that set, and the prediction was frozen before the last 13 triples were analysed.

6. *A construction for every odd $\alpha$.* The enumerated graphs without a partition consist of $\alpha - 2$ cliques and one of two small blocks of independence number 2. Gluing them along a 3-edge-connected cubic quotient gives, for every odd $\alpha \geq 3$ and $3\alpha < n \leq 4\alpha - 1$, a 3-connected graph with $t(n, \alpha) + \lceil 3\alpha/2 \rceil$ edges, independence number $\alpha$ and clique cover number $\alpha + 1$ (Theorem 14). It is extremal, without condition, at the nine such triples with $n \leq 24$, and at all of them if Didin's Theorem 9 holds.

**Organisation.** Section 2 states the prior results we use. Section 3 derives the open window. Section 4 describes the search and proves it sound and complete. Section 5 reports the validation. Section 6 gives the values and the cost. Section 7 studies the extremal graphs, and Section 8 builds them for every odd $\alpha$. Section 9 discusses limitations and open questions, Section 10 concludes, and Section 11 gives the data and the commands.

## 2 Prior results

**Bougard and Joret.** A *tree-linking* of a family of graphs adds edges between them along a tree; a *cycle-linking* adds them along a cycle. For $k \in \{1, 2\}$ Bougard and Joret determined $f$ and all extremal graphs: $f(n, \alpha, 1) = t(n, \alpha) + \alpha - 1$ for $n \geq \alpha + 1$ [2, Proposition 5], and $f(n, \alpha, 2) = t(n, \alpha) + \alpha$ for $n \geq 2\alpha + 1$ outside two exceptional cases [2, Proposition 6]. The step to $k \geq 3$ rests on two partition statements.

- [2, Proposition 7]: a connected graph $G$ with $n \geq 3\alpha$ and $\mathrm{e}(G) \leq t(n, \alpha) + \lfloor n/\alpha \rfloor + \alpha - 4$, where $\alpha = \alpha(G)$, has a partition of $V(G)$ into $\alpha$ cliques.

- [2, Proposition 8]: a 2-connected graph $G$ with $n \geq 3\alpha \geq 9$ and $\mathrm{e}(G) \leq t(n, \alpha) + \lfloor n/\alpha \rfloor + \alpha - 3$ has a partition of $V(G)$ into $\alpha$ cliques. The bound is best possible: Bougard and Joret modify $T(n, \alpha)$ into a graph $\widetilde{T}(n, \alpha)$ by joining a new vertex to every vertex of a smallest clique and pasting a second clique onto the result along one of its edges, and every cycle-linking of $\widetilde{T}(n, \alpha)$ has one more edge and no such partition [2, pp. 10–11].

The upper bound of the second line is Bougard and Joret's remark that for $n > k\alpha$ the graph $T(n, \alpha)$ can be made $k$-connected by adding $\lceil k\alpha/2 \rceil$ edges without decreasing its independence number [2, Section 6]. We call the graphs obtained this way *Bougard–Joret graphs*: a balanced union of $\alpha$ cliques with $\lceil k\alpha/2 \rceil$ edges between them, $k$-connected and with independence number $\alpha$. Explicit examples at every triple of Table 4 are among the extremal graphs listed there.

**Das and Gupta.** For $\alpha \geq 2$ and $k \geq \max(3, \alpha)$, $f(\alpha + k, \alpha, k) = \alpha k + m(k, k - \alpha)$, where $m(p, d)$ is the least size of a $d$-connected graph on $p$ vertices [9, Theorem 3.1]; in particular $f(2k - 1, k - 1, k) = k^2 - 1$ for $k \geq 4$, which refutes the first line [9, Corollary 3.3]. They also

note that a $k$-connected graph with a partition into $\alpha$ cliques has at least $t(n,\alpha)+\lceil k\alpha/2\rceil$ edges [9, Proposition 4.2], and that a proof of the second line must either establish such a partition or bound the size without it.

**Didin.** For a graph $G$ with independence number $\alpha$, an *$\alpha$-core* is a spanning subgraph $H$ with $\alpha(H)=\alpha$ in which every edge is critical (deleting it raises the independence number). Let $p(G)$ be the largest number of components of an $\alpha$-core of $G$ and $s(G)=\alpha-p(G)$. Didin's main theorem [10, Theorem 3] states that every graph $G$ with $n\geq\max(\alpha+1,p(G))$ vertices satisfies

$$\mathrm{e}(G)\ \geq\ t(n+s(G),\alpha)-s(G), \tag{3}$$

and his Theorem 9 derives the lower bound of the second line for $b$-connected graphs, $b\geq 3$, through an edge-cut form with two tight exceptions, $\alpha=2$, $n=2b+1$ and $(b,\alpha,n)=(3,3,10)$, which he shows have 2-vertex cuts [10, Theorem 10, Lemmas 17 and 18]. The preprint does not characterise the graphs that attain Theorem 9. We use only the statements, which we did not audit; every result below that depends on them is marked as conditional.

**Didin's method.** Didin's proof of his main theorem moves a graph, step by step, towards its associated Turán graph $T(n+s,\alpha)$ by *appropriate transformations*, which preserve $\alpha$ and $n-p$ and never increase a cost $x(G)$; the theorem is the statement $x(G)\geq 0$, with equality exactly for the graphs he calls tight [10, Theorem 3]. The basic step splits a connected $\alpha$-critical graph at a vertex, adding a simple vertex for each component of the $\alpha$-core of what remains [10, Theorem 5], and a series of splits ends at a disjoint union of cliques [10, Lemma 1]. Tight graphs are rebuilt from $T(n+s,\alpha)$ by a pasting move [10, Lemma 2] that goes back to Wessel's description of edge-critical graphs with connectivity two [31]. For $b$-connected graphs he adds that the $p$ components of an $\alpha$-core are joined by at least $\lceil bp/2\rceil$ edges, which is the inequality chain used in Proposition 12 below.

**Other work.** The second line is recorded as open in [20]. Wang uses the 2-connected case of Bougard and Joret to compare balanced cycle-linkings with other 2-connected graphs by their numbers of independent sets [30]. The tree strip $n=2k$, $\alpha=k-1$, and the first interior shell $n=\alpha+k+1$ [27], and the bracket $f(2d+3,2,d+2)=d^2+4d+2$ [26] lie in the first regime. The extremal problem without connectivity is Turán's theorem [29]; Brouwer showed that a graph with independence number $\alpha$ and at most $t(n,\alpha)+\lfloor n/\alpha\rfloor-2$ edges has a partition of its vertex set into $\alpha$ cliques, when $n>2\alpha$ [4]; Bougard and Joret restate the theorem in [2, Section 5] and note that the result follows from a theorem of Hanson and Toft [17]. Bougard and Joret improve this bound for connected and 2-connected graphs (their Propositions 7 and 8), and their proofs use Brooks' theorem [3], Wessel's structure of edge-critical graphs with connectivity two [31] and bounds of Erdős and Gallai [12].

Exhaustive generation has settled other small extremal values, for instance the Ramsey number $R(4,5)=25$ [21]. Dedicated generators, such as the one for cubic graphs [1], and searches for large independent sets in cubic graphs [6] follow the same pattern: a complete generator, a pruning rule proved sound, and an exact test. The search here is an instance of that pattern, built on nauty's general generator.

# 3 The open window

**Lemma 1.** *Let $\alpha \geq 3$, $k \geq 3$ and $n > k\alpha$. If $\lfloor n/\alpha \rfloor \geq \lceil k\alpha/2 \rceil - \alpha + 2$, then $f(n,\alpha,k) = t(n,\alpha) + \lceil k\alpha/2 \rceil$.*

*Proof.* The upper bound is the construction of Section 2. For the lower bound let $G$ be $k$-connected with $n$ vertices, $\alpha(G) = \alpha$ and $\mathrm{e}(G) \leq t(n,\alpha) + \lceil k\alpha/2 \rceil - 1$. The hypothesis gives $\lceil k\alpha/2 \rceil - 1 \leq \lfloor n/\alpha \rfloor + \alpha - 3$, so $\mathrm{e}(G) \leq t(n,\alpha) + \lfloor n/\alpha \rfloor + \alpha - 3$; since $n > k\alpha \geq 3\alpha \geq 9$ and $G$ is 2-connected, Proposition 8 of [2] gives a partition of $V(G)$ into $\alpha$ cliques $C_1, \ldots, C_\alpha$, all nonempty because $\alpha(G) = \alpha$. The edges inside the cliques number $\sum_i \binom{|C_i|}{2} \geq t(n,\alpha)$, because $x \mapsto \binom{x}{2}$ is convex and the sizes sum to $n$, so the sum is least when the sizes are balanced. Each $C_i$ is a proper subset of $V(G)$, so the set $\partial C_i$ of edges with exactly one end in $C_i$ has $|\partial C_i| \geq \kappa(G) \geq k$ (the edge connectivity of a graph is at least its vertex connectivity, by Whitney's inequality $\lambda(G) \geq \kappa(G)$ [32]). Every edge between cliques lies in exactly two of the sets $\partial C_i$, so there are at least $\lceil k\alpha/2 \rceil$ of them. Hence $\mathrm{e}(G) \geq t(n,\alpha) + \lceil k\alpha/2 \rceil$, a contradiction. □

Bougard and Joret note this consequence of Proposition 8 [2, p. 12]. The *open window* is the set of triples $(n,\alpha,k)$ with $\alpha \geq 3$, $k \geq 3$, $n > k\alpha$, $n \geq 2\alpha$, $n \geq \alpha + k$ and

$$\lfloor n/\alpha \rfloor \;\leq\; \lceil k\alpha/2 \rceil - \alpha + 1. \tag{4}$$

For fixed $(\alpha,k)$ the window is the finite interval $k\alpha < n < \alpha(\lceil k\alpha/2 \rceil - \alpha + 2)$. It is nonempty for all $\alpha, k \geq 3$: at $n = k\alpha + 1$ condition (4) reads $k \leq \lceil k\alpha/2 \rceil - \alpha + 1$, and $\lceil k\alpha/2 \rceil - \alpha + 1 - k = \lceil ((k-2)(\alpha-2) - 2)/2 \rceil \geq 0$. Its length grows like $(k-2)\alpha^2/2$, so the window is infinite in total. Table 1 and Figure 1 list it by order up to $n = 30$: it has 120 triples there, and it grows roughly linearly with $n$. Didin's Theorem 9, if correct, closes it entirely.

The extent of the window grows quickly with $\alpha$. For $(\alpha,k) = (3,3)$ it is the two orders 10 and 11; for $(4,3)$ the orders 13 to 15; for $(3,4)$ the orders 13 and 14; for $(3,5)$ the orders 16 to 20; for $(5,3)$ the orders 16 to 24, so that this pair is closed by the present enumeration. For $(7,3)$ the window runs from 22 to 41, and only its first three orders are decided here; in general its length is

$$\alpha\Big(\left\lceil \frac{k\alpha}{2} \right\rceil - \alpha + 2\Big) - k\alpha - 1, \tag{5}$$

which is of order $(k-2)\alpha^2/2$. The window is therefore not a finite list of exceptions that a computation can exhaust: every pair $(\alpha,k)$ contributes, and the pairs with larger $\alpha$ contribute longer intervals. For seven pairs the window ends by order 24 (Corollary 9). What a computation can do is decide its first orders, test the conjecture where the margin between Proposition 8 and the conjectured value is widest relative to the order, and recover the extremal graphs there, which is the purpose of the following sections.

# 4 Method

Figure 2 shows the pieces and how they connect.

## 4.1 A finite decision procedure

**Proposition 2.** *Let $(n,\alpha,k)$ be in the second regime, and let*

$$c = t(n,\alpha) + \left\lceil \frac{k\alpha}{2} \right\rceil, \qquad \ell = \max\Big(\left\lceil \frac{nk}{2} \right\rceil,\; t(n,\alpha)\Big). \tag{6}$$

Table 1: The open window by order $n$: the number of open-window triples, the number decided in this paper, and the pairs $(\alpha, k)$. Every triple with $n \leq 24$ is decided except where the last column of Table 3 gives a bound.

| $n$ | open | decided | pairs $(\alpha, k)$ |
|---|---|---|---|
| 10 | 1 | 1 | $(3,3)$ |
| 11 | 1 | 1 | $(3,3)$ |
| 13 | 2 | 2 | $(3,4)$, $(4,3)$ |
| 14 | 2 | 2 | $(3,4)$, $(4,3)$ |
| 15 | 1 | 1 | $(4,3)$ |
| 16 | 2 | 2 | $(3,5)$, $(5,3)$ |
| 17 | 3 | 3 | $(3,5)$, $(4,4)$, $(5,3)$ |
| 18 | 3 | 3 | $(3,5)$, $(4,4)$, $(5,3)$ |
| 19 | 5 | 5 | $(3,5)$, $(3,6)$, $(4,4)$, $(5,3)$, $(6,3)$ |
| 20 | 5 | 5 | $(3,5)$, $(3,6)$, $(4,4)$, $(5,3)$, $(6,3)$ |
| 21 | 6 | 6 | $(3,6)$, $(4,4)$, $(4,5)$, $(5,3)$, $(5,4)$, $(6,3)$ |
| 22 | 8 | 8 | $(3,6)$, $(3,7)$, $(4,4)$, $(4,5)$, $(5,3)$, $(5,4)$, $(6,3)$, $(7,3)$ |
| 23 | 8 | 8 | $(3,6)$, $(3,7)$, $(4,4)$, $(4,5)$, $(5,3)$, $(5,4)$, $(6,3)$, $(7,3)$ |
| 24 | 6 | 6 | $(3,7)$, $(4,5)$, $(5,3)$, $(5,4)$, $(6,3)$, $(7,3)$ |
| 25 | 9 | 0 | $(3,7)$, $(3,8)$, $(4,5)$, $(4,6)$, $(5,4)$, $(6,3)$, $(6,4)$, $(7,3)$, $(8,3)$ |
| 26 | 10 | 0 | $(3,7)$, $(3,8)$, $(4,5)$, $(4,6)$, $(5,4)$, $(5,5)$, $(6,3)$, $(6,4)$, $(7,3)$, $(8,3)$ |
| 27 | 10 | 0 | $(3,7)$, $(3,8)$, $(4,5)$, $(4,6)$, $(5,4)$, $(5,5)$, $(6,3)$, $(6,4)$, $(7,3)$, $(8,3)$ |
| 28 | 12 | 0 | $(3,7)$, $(3,8)$, $(3,9)$, $(4,5)$, $(4,6)$, $(5,4)$, $(5,5)$, $(6,3)$, $(6,4)$, $(7,3)$, $(8,3)$, $(9,3)$ |
| 29 | 14 | 0 | $(3,7)$, $(3,8)$, $(3,9)$, $(4,5)$, $(4,6)$, $(4,7)$, $(5,4)$, $(5,5)$, $(6,3)$, $(6,4)$, $(7,3)$, $(7,4)$, $(8,3)$, $(9,3)$ |
| 30 | 12 | 0 | $(3,8)$, $(3,9)$, $(4,5)$, $(4,6)$, $(4,7)$, $(5,4)$, $(5,5)$, $(6,4)$, $(7,3)$, $(7,4)$, $(8,3)$, $(9,3)$ |

*Then $f(n, \alpha, k) = c$ if and only if, for every $E \in \{\ell, \dots, c-1\}$, no graph exists with $n$ vertices, $E$ edges, minimum degree at least $k$, independence number exactly $\alpha$ and connectivity at least $k$. Otherwise $f(n, \alpha, k)$ is the least such $E$.*

*Proof.* A $k$-connected graph has minimum degree at least $k$, hence at least $\lceil nk/2 \rceil$ edges; a graph with independence number $\alpha$ has at least $t(n, \alpha)$ edges by Turán's theorem. So every qualifying graph has at least $\ell$ edges, and the construction of Section 2 gives one with $c$ edges. The least qualifying size is therefore either some $E \in [\ell, c-1]$ or $c$. □

The number of edge counts to search is $c - \ell$, at most $\lceil k\alpha/2 \rceil$; it ranges from one, at $(13, 4, 3)$ and $(19, 6, 3)$, to eleven, at $(24, 3, 7)$, in the window up to order 24 (Table 3). Proposition 2 turns the conjecture at one triple into finitely many existence questions, one per edge count, each answered by a complete list of candidate graphs. A negative answer at every count below $c$ is a proof that $f = c$, given the completeness of that list; no sampling or heuristic enters. The count $E = c$ itself is not needed for the value, but the same pipeline run at $E = c$ lists the extremal graphs (Section 7).

### 4.2 Generation

The candidate graphs come from `geng` of nauty 2.8.9 [23], which generates one representative of every isomorphism class of graphs with given order, by canonical construction path [22]. We call

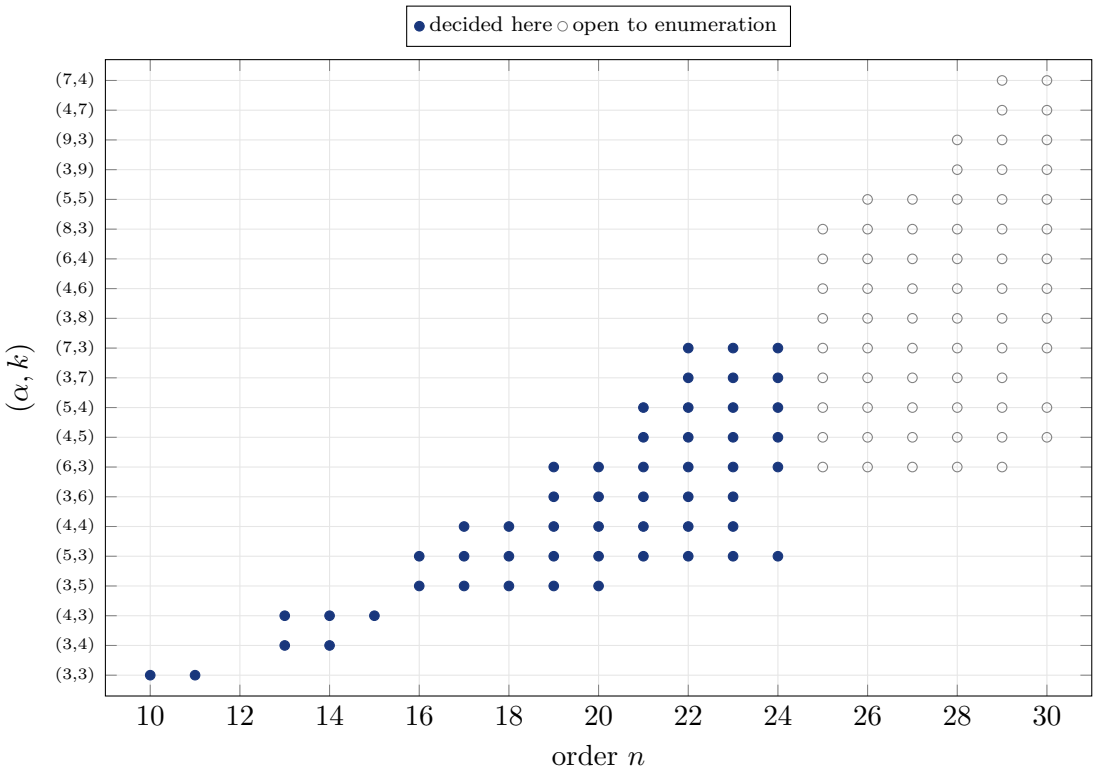



Figure 1: The open window up to order 30. Each mark is a triple $(n, \alpha, k)$ with $n > k\alpha$ outside the range of Lemma 1; filled marks are decided in this paper. For fixed $(\alpha, k)$ the window is an interval of orders starting at $k\alpha + 1$.

it as `geng -q -d`$k$ $n$ $E$`:`$E$ $r$`/`$M$: `-d`$k$ restricts the output to minimum degree at least $k$, $E$`:`$E$ to exactly $E$ edges, and $r$`/`$M$ to the $r$-th of $M$ disjoint parts. `geng` builds a graph by adding vertices $0, 1, 2, \ldots$ in order; in the words of its documentation, "each graph in the sequence is an induced subgraph of all later graphs in the sequence" (`geng.c`, version 3.7). The parts are formed at a fixed splitting level of the generation tree, where each node is assigned to one of $M$ classes and only the descendants of class $r$ are expanded; the union over $r = 0, \ldots, M-1$ is the complete output, each graph appearing in exactly one part. Gate D of Section 5 confirms this at production parameters. The number $M$ affects only the running time: the parts are disjoint and their union is the whole output, so no value depends on $M$. Many small parts balance the load across processes, at the price of repeating in every part the work above the splitting level.

### 4.3 The independence prune

`geng` accepts a compile-time hook `PRUNE(g,n,maxn)`, called for every intermediate and final graph; a nonzero return discards the graph and its whole subtree.

**Lemma 3.** *Let* $\mathrm{prune}_A(g)$ *return* 1 *exactly when the graph* $g$ *has an independent set of* $A + 1$ *vertices. Adding this hook to* `geng` *with* $A = \alpha$ *removes from the output exactly the graphs with independence number greater than* $\alpha$*.*

*Proof.* Every intermediate graph on vertices $0, \ldots, j-1$ is an induced subgraph of each of its descendants. An independent set of an induced subgraph is independent in the whole graph,

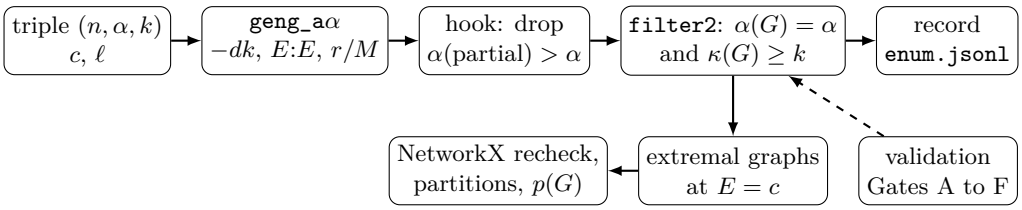



Figure 2: The search and its checks. For each edge count $E$ from $\ell$ to $c-1$, $M$ parts of the pruned generator stream into the exact filter; a kept graph would decide $f = E$. The extremal graphs at $E = c$ are regenerated and analysed by code independent of the C programs, and Gates A to F test every component at the production parameters.

**Algorithm 1:** Independence number with early stop (`filter2.c`, function `mis`)

**Input:** adjacency masks $N(v)$, candidate set $C$, current size $s$; global best $b$, cap $\alpha + 1$

```
1 if b ≥ α + 1 then return;
2 if C = ∅ then b ← max(b, s); return;
3 if s + |C| ≤ b then return ;                    // bound: cannot beat b
4 ;
5 v ← lowest vertex of C;
6 MIS(C \ ({v} ∪ N(v)), s + 1);                   // take v
7 MIS(C \ {v}, s);                                // skip v
```

so $\alpha$ never decreases along a generation path. If the hook fires at an intermediate graph, every descendant has independence number greater than $\alpha$ and would be rejected by the filter anyway; if a final graph has independence number at most $\alpha$, no ancestor has an independent set of size $\alpha + 1$ and the hook never fires on its path. The final graphs with independence number greater than $\alpha$ are discarded by the call on themselves. □

The hook is a branch-and-bound search for an independent set of size $A + 1$ on 64-bit masks (`prune_alpha.c`, Section 11). It cuts the search drastically when $\alpha$ is small relative to $n$: at $(17, 5, 3)$ with 27 edges the plain generator produces 35,329,226 graphs in 1,040 seconds, and the pruned one 482 graphs in 0.4 seconds, with the same 0 qualifying graphs.

### 4.4 The exact filter

Each generated graph, read as a `graph6` line, is kept exactly when its independence number equals $\alpha$ and its connectivity is at least $k$. The independence number is computed by Algorithm 1, an exact branch-and-bound search on 32-bit vertex masks in the spirit of the maximum-clique algorithms of Carraghan and Pardalos [7] and Östergård [25], applied to the complement implicitly. It stops as soon as an independent set of size $\alpha + 1$ is found, because the graph is then rejected regardless of the exact value.

**Lemma 4.** *Started with $C = V(G)$, $s = 0$, $b = 0$, Algorithm 1 ends with $b = \alpha(G)$ when $\alpha(G) \leq \alpha$, and with $b \geq \alpha + 1$ otherwise. Hence the test $b = \alpha$ decides $\alpha(G) = \alpha$ exactly.*

*Proof.* The two branches partition the independent sets contained in $C \cup$ (the current set) into those containing $v$ and those not containing $v$, so without the cuts the recursion visits every maximal independent set and $b$ ends at $\alpha(G)$. The bound cut discards only branches whose every

**Algorithm 2:** $k$-connectivity by exhaustive deletion (`filter2.c`, function `kconnected`)

**Input:** adjacency masks of $G$ on $n \le 32$ vertices, $k \le 9$

**if** $n \le k$ **then return** false;
**if** *$G$ is not connected* **then return** false;
**for** $j = 1$ **to** $k - 1$ **do**
  **foreach** *$j$-subset $S \subseteq V(G)$ in lexicographic order* **do**
    **if** *$G - S$ is not connected (breadth-first search on masks)* **then return** false;
**return** true;

**Algorithm 3:** Deciding $f(n, \alpha, k)$ on an open-window triple (`run_enum.py`)

**Input:** $(n, \alpha, k)$; number of parts $M$; number of processes $J$; optional time cap per edge count

$c \leftarrow t(n, \alpha) + \lceil k\alpha/2 \rceil$; $\ell \leftarrow \max(\lceil nk/2 \rceil, t(n, \alpha))$;
**for** $E = \ell$ **to** $c - 1$ **do**
  run, on $J$ processes, the $M$ pipes `geng_a`$\alpha$ `-q -d`$k$ $n$ $E$`:`$E$ $r$`/`$M$ `| filter2` $\alpha$ $k$;
  record the number of generated graphs, kept graphs, the first kept examples and the wall-clock time;
  **if** *a graph is kept* **then return** $f = E$;
  **if** *the cap is reached* **then return** $f \ge E$;
**return** $f = c$;

completion has at most $b$ vertices, so it never discards a strictly larger set. The early stop fires only once $b \ge \alpha + 1$, and before that the search is exact. So a graph with $\alpha(G) > \alpha$ ends with $b \ge \alpha + 1$; the value may exceed $\alpha + 1$, because $b$ is updated only at the leaves of the recursion, where the whole current set is counted. A graph with $\alpha(G) \le \alpha$ never triggers the stop and ends with $b = \alpha(G)$. The filter keeps a graph only when $b = \alpha$. □

Connectivity is tested by Algorithm 2: a graph with more than $k$ vertices is $k$-connected exactly when it stays connected after the deletion of every vertex set of size at most $k - 1$ (the definition of vertex connectivity). Each test is a breadth-first search on masks. The number of deleted sets is $\sum_{j=0}^{k-1} \binom{n}{j}$, at most 190,051 for $n = 24$, $k = 7$; the test runs only on graphs whose independence number is exactly $\alpha$, a small fraction of the stream. Asymptotically faster algorithms exist [14, 13]; the exhaustive one is preferred here because its correctness is immediate.

**Lemma 5.** *Algorithm 2 returns true exactly when $\kappa(G) \ge k$.*

*Proof.* By definition $\kappa(G) \ge k$ means $n > k$ and no set of fewer than $k$ vertices disconnects $G$; the algorithm tests exactly these sets, including the empty one. □

### 4.5 The decision procedure

Algorithm 3 combines the pieces. Each part is a pipe from the pruned generator into the filter; parts run in parallel, and a triple's value is decided by the first edge count with a kept graph, or by the exhaustion of all counts below $c$.

**Theorem 6** (soundness)**.** *Assume that* `geng` *outputs one representative of every isomorphism class it is asked for, and that its split into $M$ parts partitions that output. If Algorithm 3 returns $f = c$, then $f(n, \alpha, k) = c$; if it returns $f = E < c$, then $f(n, \alpha, k) = E$.*

*Proof.* By the assumption, which Gates D and E test but do not prove, the generator without the hook outputs one representative of every isomorphism class with $n$ vertices, $E$ edges and minimum degree at least $k$. By Lemma 3 the hook removes only graphs with independence number greater than $\alpha$. By Lemmas 4 and 5 the filter keeps exactly the graphs with independence number $\alpha$ and connectivity at least $k$. The claim follows from Proposition 2. □

Algorithm 3 ran with $M = 56$ parts on $J = 28$ processes from EXP-007 on; EXP-005 and the first runs of EXP-006 used the generator without the hook and $M \in \{48, 96, 224, 448, 896, 1792\}$. The decisions recorded for a triple use the pruned generator whenever both exist; where both exist they agree on the number of kept graphs at every edge count (Section 5). A cap of four hours per edge count applied at order 24 in EXP-009; EXP-010 reran the two capped counts with a cap of twenty hours.

### 4.6 Cost

The time of one edge count is, to first order,

$$T(E) \;\approx\; N(E)\,(\tau_{\mathrm{gen}} + \tau_{\alpha}) \;+\; N_{\alpha}(E)\,\tau_{\mathrm{bfs}} \sum_{j=0}^{k-1} \binom{n}{j}, \tag{7}$$

where $N(E)$ is the number of graphs the pruned generator outputs, $N_{\alpha}(E)$ the number among them with independence number exactly $\alpha$, $\tau_{\mathrm{gen}}$ and $\tau_{\alpha}$ the per-graph costs of generation and of Algorithm 1, and $\tau_{\mathrm{bfs}}$ the cost of one breadth-first search. The hook removes every graph with independence number above $\alpha$, and Turán's theorem removes most of those below it: a graph with independence number at most $\alpha - 1$ has at least $t(n, \alpha - 1)$ edges, so

$$t(n, \alpha - 1) \;\geq\; c \quad\Longrightarrow\quad N_{\alpha}(E) = N(E) \;\text{ for every } E < c. \tag{8}$$

The condition holds at 44 of the 53 triples. There the independence test never rejects, and the filter's work is the connectivity test; Gate B observes $N_{\alpha} = N$ on every sampled slice, including the triples with $\alpha = 7$, where the condition fails. The connectivity test dominates for large $k$: at $n = 24$, $k = 7$ the sum in (7) is 190,051, against $1 + 24 + 276 = 301$ at $k = 3$. The per-graph generation cost $\tau_{\mathrm{gen}}$ is not a constant either: when $k$ is large, most partial graphs admit no completion with minimum degree $k$, and the generator spends most of its time on them, so the cost per output graph grows (Section 6 gives the case of $(22, 3, 7)$).

### 4.7 Structure of the extremal graphs

For the extremal graphs (those with $c$ edges) we regenerate the complete list at $E = c$ with the same pipeline, keeping every qualifying `graph6` line, and analyse each graph with code that shares nothing with the C programs (NetworkX 3.4.1 [16]). Three quantities are computed exactly.

- *Recheck.* The edge count, the independence number as a maximum clique of the complement, and the vertex connectivity by NetworkX's flow-based routine.

- *Clique partition.* Whether $V(G)$ splits into $\alpha$ cliques, by an exact backtracking $\alpha$-colouring of the complement with the usual symmetry breaking (a new colour only as the next unused one); and, independently, whether $\alpha$ maximal cliques cover $V(G)$, by listing the maximal cliques (Bron–Kerbosch with pivoting [5, 28]) and searching their $\alpha$-subsets with mask unions. Both return the clique cover number $\theta(G)$ when it is at most $\alpha + 2$.
- *Didin's core parameter.* $p(G)$ by Lemma 7 below and $s(G) = \alpha - p(G)$.

**Lemma 7.** *$p(G)$ is the largest $q$ for which $V(G)$ has a partition into $q$ nonempty parts $V_1, \dots, V_q$ with $\sum_i \alpha(G[V_i]) \le \alpha(G)$. In particular $p(G) = \alpha(G)$ exactly when $V(G)$ splits into $\alpha(G)$ cliques, that is, when $\theta(G) = \alpha(G)$.*

*Proof.* For every partition, $\alpha(G) \le \sum_i \alpha(G[V_i])$, since an independent set of $G$ meets each part in an independent set. If equality holds, the disjoint union $G'$ of the $G[V_i]$ is a spanning subgraph with $\alpha(G') = \alpha(G)$; deleting non-critical edges of $G'$ one at a time keeps the independence number and ends at an $\alpha$-core of $G$ whose components refine the partition, so $p(G) \ge q$. Conversely, the components of an $\alpha$-core $H$ partition $V(G)$, and $\sum \alpha(\text{component}) = \alpha(H) = \alpha(G)$, while $\alpha(G[V_i]) \le \alpha(H[V_i])$ because $H[V_i]$ is a spanning subgraph of $G[V_i]$. For the last claim, a part has $\alpha(G[V_i]) \ge 1$, so $q \le \alpha$, and $q = \alpha$ forces every part to have independence number one, that is, to be a clique. □

The search for $p(G)$ assigns vertices in order to at most $q$ parts, opens a new part only as the next index, keeps each part's independence number exactly (memoised on masks) and cuts a branch as soon as the sum exceeds $\alpha$; it decides each $q$ from $\alpha$ down. On 400 random graphs with 4 to 9 vertices it agrees with a brute force over all set partitions.

## 5 Validation

Table 2 lists every check, its scope and its outcome. We summarise the ones that bear on the production parameters.

The gates target distinct ways the search could go wrong. Gate F checks that the binaries that ran are the committed sources. Gates A and B check that the filter's two tests are exact at the parameters used, including the connectivity test at $k = 6, 7$, which the earlier check at order 8 never reached. Gate C checks that the hook rejects exactly the graphs it should, the only way it could lose a qualifying graph. Gate D checks that splitting the generator into parts loses or duplicates nothing when the hook is active. Gate E checks the generator itself at orders beyond the earlier checks. Two defects of the validation harness, not of the search, were found and fixed on the way. Parallel parts writing into one pipe can interleave lines, which made a first Gate D run report different hashes for equal counts; with one file per part the outputs agree byte for byte. A quoting error emptied the first Gate E counts. The records of both first runs are kept with the experiment.

- *Gate F, provenance.* The production filter rebuilt from its committed source with the recorded build line is byte-identical to the binary that ran (SHA-256 `ebbfd34da19290f5...`). The five pruned generators rebuilt from nauty 2.8.9's `geng.c` (SHA-256 `6dda2361ade7714f...`) and the committed hook with nauty's makefile flags (`-O4 -mpopcnt -march=native`) are byte-identical to the production ones. The build line written in the hook's source header uses `-O3` without `-mpopcnt` and `-march=native` and gives other bytes; Section 11 states the line that reproduces the binaries.

- *Gate A, filter on graphs with known answers.* 3,519 graphs of orders 10 to 24 from seven families (Harary graphs $H_{k,n}$ [19], circulants, random regular graphs, complements of sparse random graphs, unions of cliques with random cross edges, random graphs, and the extremal graphs of Section 7), with independence numbers from 1 to 12 and connectivity from 0 to 22. Each graph is decided at every $(a, k') \in \{\alpha-1, \alpha, \alpha+1\} \times \{\kappa-1, \kappa, \kappa+1\}$ with $1 \le k' \le 9$: 26,005 decisions in 105 groups, 5,703 of them keeps, and no disagreement with NetworkX.

- *Gate B, filter on production streams.* For each production triple with $k \ge 6$ or $\alpha = 7$, slices of the pruned generator's output at $E \in \{c-1, c, c+1\}$, at most 3,000 graphs each: 21,196 graphs in 33 slices, no graph with independence number above the bound, and no disagreement. A slice is a fixed part of the generation tree, so 19 slices are empty; the nonempty ones are $(19, 3, 6)$ at $E = 59, 60, 61$; $(20, 3, 6)$ at $E = 65, 66$; $(21, 3, 6)$ at $E = 71$; $(22, 3, 6)$ at $E = 78, 79, 80$; $(22, 3, 7)$ at $E = 82$; $(22, 7, 3)$ at $E = 34, 35$; $(23, 7, 3)$ at $E = 37, 39$. No slice at order 24 and no production edge count with $k = 7$ is covered, so the filter at those parameters rests on its proofs and on Gate A.

- *Gate C, the hook alone.* A harness calls the production hook on whole graphs at bounds $A = 2, \ldots, 8$: 3,876 graphs, 27,132 decisions, no disagreement with NetworkX.

- *Gate D, the split under the hook.* For $(17, 5, 3)$ at $E = 29$ (218,377 graphs), $(19, 3, 6)$ at $E = 60$ (880 graphs), $(20, 4, 4)$ at $E = 48$ (266,577 graphs), the sorted output of the pruned generator in one part equals the union of its 56 parts, byte for byte (equal SHA-256); 0 differences.

- *Gate E, the generator's counts.* `geng` reproduces all 10 values checked against the OEIS [24]: connected graphs of orders 9 and 10 (A001349), connected cubic graphs of orders 16 to 22 (A002851) and connected quartic graphs of orders 12 to 15 (A006820), the largest count being 11,716,571; 0 differences.

The families of Gate A are chosen to stress the two tests where they could fail. Harary graphs attain the least number of edges for a given connectivity, so they sit exactly at $\kappa = k$, where an off-by-one error in the deletion loop would show. Unions of cliques with random cross edges have the shape of the extremal graphs, with small $\alpha$ and connectivity decided by a few cross edges. Complements of sparse random graphs give small independence numbers with high connectivity, up to 22, beyond any production value. Random regular graphs and circulants give every connectivity up to their degree, and random graphs of every density cover what remains. The decisions at $a = \alpha \pm 1$ and $k' = \kappa \pm 1$ test both sides of each threshold.

## 6 Values and cost

**Theorem 8.** *For every open-window triple $(n, \alpha, k)$ with $n \le 24$, $f(n, \alpha, k) = t(n, \alpha) + \lceil k\alpha/2 \rceil$.*

**Corollary 9.** *For each of the 7 pairs $(\alpha, k) \in \{(3,3), (4,3), (5,3), (3,4), (4,4), (3,5), (3,6)\}$, $f(n, \alpha, k) = t(n, \alpha) + \lceil k\alpha/2 \rceil$ for every $n > k\alpha$.*

*Proof.* For these pairs the whole open window, the orders $k\alpha < n < \alpha(\lceil k\alpha/2 \rceil - \alpha + 2)$, has $n \le 24$ and is covered by Theorem 8; the larger orders are covered by Lemma 1. □

For these pairs the second line is settled independently of Didin's argument, by Bougard and Joret's Proposition 8 above the window and by exhaustive enumeration inside it. The next pairs

Table 2: Validation of the search: each check, the experiment that holds its record, its scope, the number of items checked and the outcome. Independent implementation: NetworkX 3.4.1. Every row passed.

| check | record | scope | items | outcome |
|---|---|---|---|---|
| Generator counts | EXP-005 | connected graphs, $n = 5, 7, 8$ | 3 | A001349: 21, 853, 11,117 |
| Cubic graphs | EXP-005 | $(16, 5, 3)$, $E = 24$ | 4,207 | A005638(8) |
| Known values | EXP-005, 006 | $f(7, 3, 4) = 15$, $f(12, 3, 3) = 23$ | 2 | reproduced |
| Filter vs. NetworkX | EXP-006 | all graphs, $n = 8$, $\alpha \le 6$, $k \le 5$ | 12,346 | 0 mismatches |
| Two filters | EXP-006 | 4 edge counts, $n \le 16$ | 27,235,386 | kept counts equal |
| Prune end to end | EXP-006 | 12 edge counts, $n \le 17$ | 12 | kept counts equal |
| Plain vs. pruned | EXP-006 | $(16, 5, 3)$, $(17, 5, 3)$ | 4 | kept counts equal |
| F: provenance | EXP-012 | filter2, geng_a3 to geng_a7 | 6 binaries | byte-identical |
| A: filter, known answers | EXP-012 | $n \le 24$, $k' \le 9$, $\alpha \le 12$ | 26,005 | 0 mismatches |
| B: filter, production | EXP-012 | $k \ge 6$ or $\alpha = 7$, $E = c - 1, c, c + 1$ | 21,196 | 0 mismatches |
| C: prune hook | EXP-012 | bounds 2 to 8, $n \le 24$ | 27,132 | 0 mismatches |
| D: split under the hook | EXP-012 | 3 triples, mod 1 vs. 56 | 3 | 0 differences |
| E: generator counts | EXP-012 | A001349, A002851, A006820 | 10 | 0 differences |
| Extremal recount | EXP-011 | 35 triples at $E = c$ | 246 | 0 differences from collect |
| Extremal recheck | EXP-011 | edges, $\alpha$, $\kappa$ by NetworkX | 246 | 0 failures |
| Two partition methods | EXP-011 | colouring vs. maximal cliques | 246 | 0 disagreements |
| Core search | EXP-011 | vs. all set partitions, $n \le 9$ | 400 | 0 differences |

whose windows are only partly covered are $(6, 3)$, $(4, 5)$, $(5, 4)$, $(3, 7)$ and $(7, 3)$, decided up to order 24.

Table 3 lists every open-window triple with $n \le 24$: $t(n, \alpha)$, the conjectured value $c$, the value found, the edge counts searched, the graphs examined below $c$, the wall-clock seconds, and, where the enumeration also covered $E = c$, the number of extremal graphs. Appendix A lists every enumeration record, one row per triple, edge count and run.

All 53 values equal the conjectured ones. The two order-24 edge counts left open by the first pass at order 24 were the largest single computations: 23,556,793 graphs at $(24, 7, 3)$ with $E = 40$, in 11.9 hours on 14 processes, and 569,264 graphs at $(24, 3, 7)$ with $E = 94$, in 10.8 hours. Almost all of the plain generator's share of the total comes from $(14, 3, 4)$ and $(15, 4, 3)$ in the first experiment, before the hook existed; with the hook, no single edge count produced more than a few tens of millions of graphs.

Table 3: Values on the open window with $n \le 24$. $f$: the value found; $\ge E$: no qualifying graph below $E$, and the count $E$ was not completed. Graphs: candidates examined below $c$ (pruned generator where available). Seconds: wall-clock time on 28 processes. At $c$: the number of extremal graphs when $E = c$ was enumerated.

| $(n, \alpha, k)$ | $t(n, \alpha)$ | $c$ | $f$ | edges searched | graphs | seconds | at $c$ |
|---|---|---|---|---|---|---|---|
| $(10, 3, 3)$ | 12 | 17 | 17 | 15–16 | 383 | 0 | 15 |
| $(11, 3, 3)$ | 15 | 20 | 20 | 17–19 | 55,812 | 0 | 6 |
| $(13, 3, 4)$ | 22 | 28 | 28 | 26–27 | 1,095,032 | 2 | 1 |
| $(13, 4, 3)$ | 15 | 21 | 21 | 20–20 | 1,999 | 0 | 1 |
| $(14, 3, 4)$ | 26 | 32 | 32 | 28–31 | 5,268,961,461 | 754 | – |
| $(14, 4, 3)$ | 18 | 24 | 24 | 21–23 | 2,147,042 | 4 | 1 |
| $(15, 4, 3)$ | 21 | 27 | 27 | 23–26 | 904,211,641 | 284 | 1 |
| $(16, 3, 5)$ | 35 | 43 | 43 | 40–42 | 134 | 0 | 3 |

| $(n,\alpha,k)$ | $t(n,\alpha)$ | $c$ | $f$ | edges searched | graphs | seconds | at $c$ |
|---|---|---|---|---|---|---|---|
| $(16,5,3)$ | 18 | 26 | 26 | 24–25 | 544 | 0 | 18 |
| $(17,3,5)$ | 40 | 48 | 48 | 43–47 | 2,120 | 4 | 3 |
| $(17,4,4)$ | 28 | 36 | 36 | 34–35 | 99 | 1 | 2 |
| $(17,5,3)$ | 21 | 29 | 29 | 26–28 | 11,780 | 2 | 34 |
| $(18,3,5)$ | 45 | 53 | 53 | 45–52 | 6,568 | 9 | 2 |
| $(18,4,4)$ | 32 | 40 | 40 | 36–39 | 4,086 | 9 | 4 |
| $(18,5,3)$ | 24 | 32 | 32 | 27–31 | 71,021 | 10 | 35 |
| $(19,3,5)$ | 51 | 59 | 59 | 51–58 | 17,920 | 26 | 4 |
| $(19,3,6)$ | 51 | 60 | 60 | 57–59 | 83 | 9 | 1 |
| $(19,4,4)$ | 36 | 44 | 44 | 38–43 | 24,673 | 19 | 2 |
| $(19,5,3)$ | 27 | 35 | 35 | 29–34 | 149,597 | 9 | 18 |
| $(19,6,3)$ | 21 | 30 | 30 | 29–29 | 295 | 2 | 2 |
| $(20,3,5)$ | 57 | 65 | 65 | 57–64 | 21,372 | 52 | 4 |
| $(20,3,6)$ | 57 | 66 | 66 | 60–65 | 4,046 | 104 | 1 |
| $(20,4,4)$ | 40 | 48 | 48 | 40–47 | 41,808 | 22 | 2 |
| $(20,5,3)$ | 30 | 38 | 38 | 30–37 | 132,654 | 18 | 5 |
| $(20,6,3)$ | 24 | 33 | 33 | 30–32 | 23,316 | 29 | 5 |
| $(21,3,6)$ | 63 | 72 | 72 | 63–71 | 29,091 | 342 | 1 |
| $(21,4,4)$ | 45 | 53 | 53 | 45–52 | 141,850 | 113 | 2 |
| $(21,4,5)$ | 45 | 55 | 55 | 53–54 | 420 | 232 | 1 |
| $(21,5,3)$ | 34 | 42 | 42 | 34–41 | 471,911 | 105 | 17 |
| $(21,5,4)$ | 34 | 44 | 44 | 42–43 | 1,298 | 204 | 7 |
| $(21,6,3)$ | 27 | 36 | 36 | 32–35 | 303,615 | 93 | 5 |
| $(22,3,6)$ | 70 | 79 | 79 | 70–78 | 83,931 | 633 | 1 |
| $(22,3,7)$ | 70 | 81 | 81 | 77–80 | 782 | 1,716 | – |
| $(22,4,4)$ | 50 | 58 | 58 | 50–57 | 222,379 | 303 | 4 |
| $(22,4,5)$ | 50 | 60 | 60 | 55–59 | 47,945 | 2,379 | – |
| $(22,5,3)$ | 38 | 46 | 46 | 38–45 | 874,494 | 290 | 31 |
| $(22,5,4)$ | 38 | 48 | 48 | 44–47 | 101,779 | 2,061 | – |
| $(22,6,3)$ | 30 | 39 | 39 | 33–38 | 1,365,978 | 113 | 5 |
| $(22,7,3)$ | 24 | 35 | 35 | 33–34 | 20,112 | 800 | – |
| $(23,3,6)$ | 77 | 86 | 86 | 77–85 | 100,793 | 1,306 | – |
| $(23,3,7)$ | 77 | 88 | 88 | 81–87 | 62,230 | 13,918 | – |
| $(23,4,4)$ | 55 | 63 | 63 | 55–62 | 190,804 | 342 | – |
| $(23,4,5)$ | 55 | 65 | 65 | 58–64 | 610,557 | 7,350 | – |
| $(23,5,3)$ | 42 | 50 | 50 | 42–49 | 954,295 | 200 | – |
| $(23,5,4)$ | 42 | 52 | 52 | 46–51 | 1,410,017 | 5,162 | – |
| $(23,6,3)$ | 33 | 42 | 42 | 35–41 | 2,519,045 | 241 | – |
| $(23,7,3)$ | 27 | 38 | 38 | 35–37 | 1,456,747 | 6,526 | – |
| $(24,3,7)$ | 84 | 95 | 95 | 84–94 | 708,187 | 45,162 | – |
| $(24,4,5)$ | 60 | 70 | 70 | 60–69 | 1,860,761 | 7,473 | – |
| $(24,5,3)$ | 46 | 54 | 54 | 46–53 | 644,641 | 369 | – |
| $(24,5,4)$ | 46 | 56 | 56 | 48–55 | 5,849,782 | 8,228 | – |
| $(24,6,3)$ | 36 | 45 | 45 | 36–44 | 2,062,478 | 313 | – |
| $(24,7,3)$ | 30 | 41 | 41 | 36–40 | 24,371,372 | 45,116 | – |

**Cost.** The search examined 6,223,452,780 graphs below the conjectured values: 6,176,473,370 with the plain generator in EXP-005 and EXP-006 and 46,979,410 with the pruned one. The wall-clock times of the edge counts sum to 42.4 hours on an Intel Core i9-14900HX under Ubuntu 24.04 in WSL2, with 28 processes per edge count, except the two counts of EXP-010, which ran at the

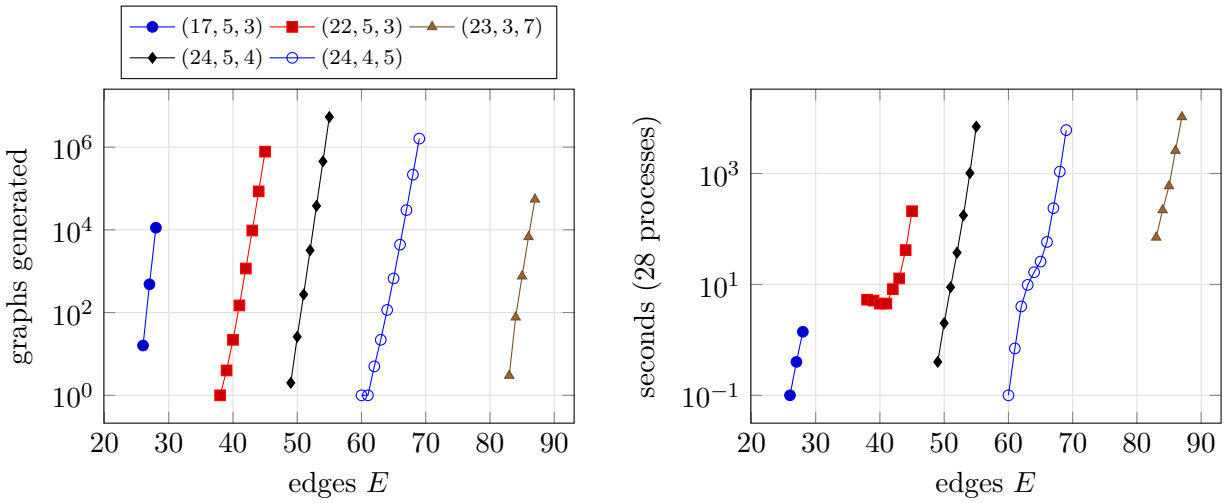



Figure 3: Candidates generated (left) and wall-clock seconds (right) per edge count below the conjectured value, for five triples, with the pruned generator. Each curve ends at $c - 1$.

same time on 14 processes each. Figure 3 shows how the number of candidates and the time grow with the edge count for five triples. Within a triple, once the counts exceed a thousand, each added edge multiplies the number of candidates by a factor between 6.8 and 11.9 on the plotted triples, and the factor itself rises slowly with $E$; the time follows the count, above a floor of a few seconds set by process start-up. Summed over the decided triples of one order, the time below $c$ was 1,088 seconds at order 21, 8,296 at order 22 and 35,045 at order 23, factors of 7.6 and 4.2; at order 24 two edge counts needed more than four hours each. The pruned generator's advantage is largest when $\alpha$ is small relative to $n/k$: the hook then fires early in the generation tree.

The cost is not spread evenly. Where $k$ is large, the generator's search dominates: at $(22, 3, 7)$ only 782 graphs were output below $c$, but the search took 1,716 seconds, spent mostly in partial graphs that admit no completion with minimum degree 7. Where $\alpha$ is large and $k = 3$, the output itself is large: $(23, 7, 3)$ produced 1,456,747 graphs in 6,526 seconds, and the last edge count of $(24, 7, 3)$ produced 23,556,793. Between these extremes, triples with $\alpha = 5$ and $k = 3$ are cheap, a few hundred seconds even at order 24, because the hook removes most partial graphs early and the filter needs only pairs and single vertices for the connectivity test.

## 7 The extremal graphs

At the 35 triples where the enumeration covered $E = c$, we regenerated every extremal graph. In every case the count matches the earlier enumeration, and every graph passes the independent recheck (edges, independence number, connectivity). Table 4 gives, per triple, the number of extremal graphs, how many split into $\alpha$ cliques, and the distribution of Didin's parameter $s$; Figure 4 shows the split.

**Lemma 10.** *Let $(n, \alpha, k)$ be in the second regime and $G$ a $k$-connected graph with $n$ vertices, independence number $\alpha$ and $t(n, \alpha) + \lceil k\alpha/2 \rceil$ edges. If $V(G)$ splits into $\alpha$ cliques $C_1, \dots, C_\alpha$, then the clique orders are $\lfloor n/\alpha \rfloor$ or $\lceil n/\alpha \rceil$, the cliques span exactly $t(n, \alpha)$ edges, and exactly $\lceil k\alpha/2 \rceil$ edges join different cliques, every clique sending at least $k$ of them.*

*Proof.* As in the proof of Lemma 1, the edges inside the cliques number $\sum_i \binom{|C_i|}{2} \geq t(n, \alpha)$ and

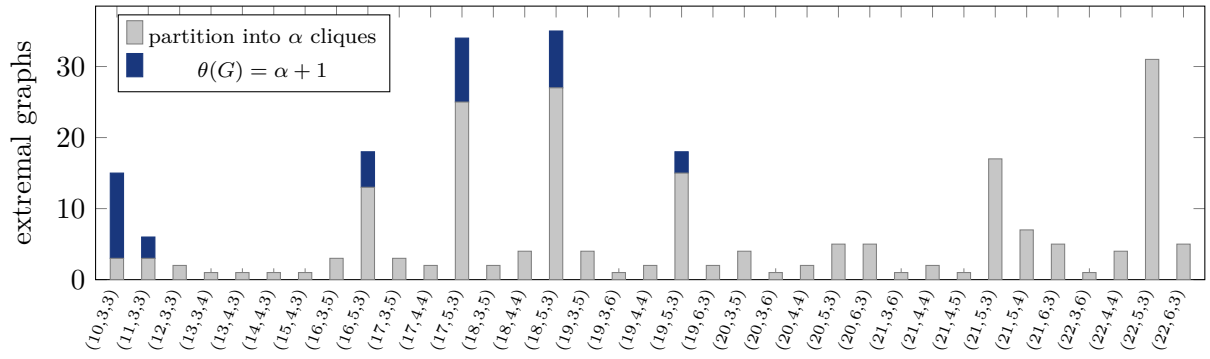



Figure 4: Extremal graphs at the conjectured value, per triple: with a partition into $\alpha$ cliques (grey, all of Bougard–Joret structure) and with clique cover number $\alpha + 1$ (dark). The dark bars occur only at $k = 3$ with $\alpha$ odd and $n \le 4\alpha - 1$.

the edges between them at least $\lceil k\alpha/2 \rceil$. Their sum is the number of edges, so both inequalities are equalities. Equality in the first holds only for balanced orders, because $\binom{x}{2}$ is strictly convex on the integers. □

So every extremal graph is either a Bougard–Joret graph, a balanced union of cliques with $\lceil k\alpha/2 \rceil$ edges between them, or a graph with $\theta(G) > \alpha$. The second kind exists: 40 of the 246 extremal graphs have no partition into $\alpha$ cliques, both methods of Section 4.7 agree on each, and all have $\theta(G) = \alpha + 1$. They occur only at $k = 3$ with $\alpha$ odd, at $(10, 3, 3)$, $(11, 3, 3)$ and $(16, 5, 3)$ to $(19, 5, 3)$; at $(20, 5, 3)$, $(21, 5, 3)$ and $(22, 5, 3)$ every extremal graph is a Bougard–Joret graph. Figure 5 draws one of each kind.

All 246 extremal graphs have connectivity exactly $k$. The Bougard–Joret graphs have automorphism groups of orders from 1 to 1,296, depending on how symmetric the arrangement of the $\lceil k\alpha/2 \rceil$ edges between the cliques is. The graphs with $s(G) = 1$ have automorphism groups of order 1 or 2, and those with $s(G) = 2$ of order at most 4; their clique numbers are 3 or 4, against 4 to 8 for the Bougard–Joret graphs.

The location of these graphs is explained by Didin's inequality (3) and the cut counting in the proof of his Theorem 10. We first record an elementary fact, which is his Lemma 19 [10].

**Lemma 11.** *For $N \ge \alpha$, $t(N + 1, \alpha) - t(N, \alpha) = \lfloor N/\alpha \rfloor$. Hence for $n > k\alpha$ and $s \ge 0$, $t(n + s, \alpha) - t(n, \alpha) = \sum_{i=0}^{s-1} \lfloor (n + i)/\alpha \rfloor \ge ks$, with equality for $s \ge 1$ exactly when $\lfloor (n + s - 1)/\alpha \rfloor = k$.*

*Proof.* Adding a vertex to $T(N, \alpha)$ in a smallest clique, of order $\lfloor N/\alpha \rfloor$, gives $T(N + 1, \alpha)$ and adds $\lfloor N/\alpha \rfloor$ edges. The terms of the sum are nondecreasing and each is at least $\lfloor n/\alpha \rfloor \ge k$. □

**Proposition 12** (conditional on [10, Theorem 3])**.** *Let $(n, \alpha, k)$ be in the second regime with $\alpha \ge 3$, and let $G$ be $k$-connected with $n$ vertices, independence number $\alpha$, $t(n, \alpha) + \lceil k\alpha/2 \rceil$ edges and no partition into $\alpha$ cliques. Then either*

*(a) $p(G) \ge 2$, $k = 3$, $\alpha$ is odd, $s(G) = 1$ and $n \le 4\alpha - 1$; or*

*(b) $p(G) = 1$ and $(n, \alpha, k)$ is one of $(10, 3, 3)$, $(11, 3, 3)$, $(13, 4, 3)$, $(16, 5, 3)$, $(13, 3, 4)$, $(16, 3, 5)$.*

*Proof.* Write $p = p(G)$, $s = s(G) \ge 1$ (Lemma 7) and $m = t(n, \alpha) + \lceil k\alpha/2 \rceil$. Let $H$ be an $\alpha$-core with $p$ components on vertex sets $V_1, \dots, V_p$. Then $H$ is $\alpha$-critical with $p(H) = p$, so (3) applied to $H$ gives $\mathrm{e}(H) \ge t(n + s, \alpha) - s$.

Table 4: The extremal graphs at the conjectured value. Graphs: all extremal graphs (each rechecked independently). Partition: those with a partition into $\alpha$ cliques, all of Bougard–Joret structure. $s$: the distribution of Didin's parameter $s(G) = \alpha - p(G)$. Allowed: the values of $s$ that Proposition 12 permits.

| triple | $c$ | graphs | recorded | partition | $\theta = \alpha + 1$ | $s$ (value:count) | allowed $s$ |
|---|---|---|---|---|---|---|---|
| $(10, 3, 3)$ | 17 | 15 | 15 | 3 | 12 | 0:3, 1:4, 2:8 | 0, 1, 2 |
| $(11, 3, 3)$ | 20 | 6 | 6 | 3 | 3 | 0:3, 1:3 | 0, 1, 2 |
| $(12, 3, 3)$ | 23 | 2 | 2 | 2 | 0 | 0:2 | 0 |
| $(13, 3, 4)$ | 28 | 1 | 1 | 1 | 0 | 0:1 | 0, 2 |
| $(13, 4, 3)$ | 21 | 1 | 1 | 1 | 0 | 0:1 | 0, 3 |
| $(14, 4, 3)$ | 24 | 1 | 1 | 1 | 0 | 0:1 | 0 |
| $(15, 4, 3)$ | 27 | 1 | 1 | 1 | 0 | 0:1 | 0 |
| $(16, 3, 5)$ | 43 | 3 | 3 | 3 | 0 | 0:3 | 0, 2 |
| $(16, 5, 3)$ | 26 | 18 | 18 | 13 | 5 | 0:13, 1:5 | 0, 1, 4 |
| $(17, 3, 5)$ | 48 | 3 | 3 | 3 | 0 | 0:3 | 0 |
| $(17, 4, 4)$ | 36 | 2 | 2 | 2 | 0 | 0:2 | 0 |
| $(17, 5, 3)$ | 29 | 34 | 34 | 25 | 9 | 0:25, 1:9 | 0, 1 |
| $(18, 3, 5)$ | 53 | 2 | 2 | 2 | 0 | 0:2 | 0 |
| $(18, 4, 4)$ | 40 | 4 | 4 | 4 | 0 | 0:4 | 0 |
| $(18, 5, 3)$ | 32 | 35 | 35 | 27 | 8 | 0:27, 1:8 | 0, 1 |
| $(19, 3, 5)$ | 59 | 4 | 4 | 4 | 0 | 0:4 | 0 |
| $(19, 3, 6)$ | 60 | 1 | 1 | 1 | 0 | 0:1 | 0 |
| $(19, 4, 4)$ | 44 | 2 | 2 | 2 | 0 | 0:2 | 0 |
| $(19, 5, 3)$ | 35 | 18 | 18 | 15 | 3 | 0:15, 1:3 | 0, 1 |
| $(19, 6, 3)$ | 30 | 2 | 2 | 2 | 0 | 0:2 | 0 |
| $(20, 3, 5)$ | 65 | 4 | 4 | 4 | 0 | 0:4 | 0 |
| $(20, 3, 6)$ | 66 | 1 | 1 | 1 | 0 | 0:1 | 0 |
| $(20, 4, 4)$ | 48 | 2 | 2 | 2 | 0 | 0:2 | 0 |
| $(20, 5, 3)$ | 38 | 5 | 5 | 5 | 0 | 0:5 | 0 |
| $(20, 6, 3)$ | 33 | 5 | 5 | 5 | 0 | 0:5 | 0 |
| $(21, 3, 6)$ | 72 | 1 | 1 | 1 | 0 | 0:1 | 0 |
| $(21, 4, 4)$ | 53 | 2 | 2 | 2 | 0 | 0:2 | 0 |
| $(21, 4, 5)$ | 55 | 1 | 1 | 1 | 0 | 0:1 | 0 |
| $(21, 5, 3)$ | 42 | 17 | 17 | 17 | 0 | 0:17 | 0 |
| $(21, 5, 4)$ | 44 | 7 | 7 | 7 | 0 | 0:7 | 0 |
| $(21, 6, 3)$ | 36 | 5 | 5 | 5 | 0 | 0:5 | 0 |
| $(22, 3, 6)$ | 79 | 1 | 1 | 1 | 0 | 0:1 | 0 |
| $(22, 4, 4)$ | 58 | 4 | 4 | 4 | 0 | 0:4 | 0 |
| $(22, 5, 3)$ | 46 | 31 | 31 | 31 | 0 | 0:31 | 0 |
| $(22, 6, 3)$ | 39 | 5 | 5 | 5 | 0 | 0:5 | 0 |

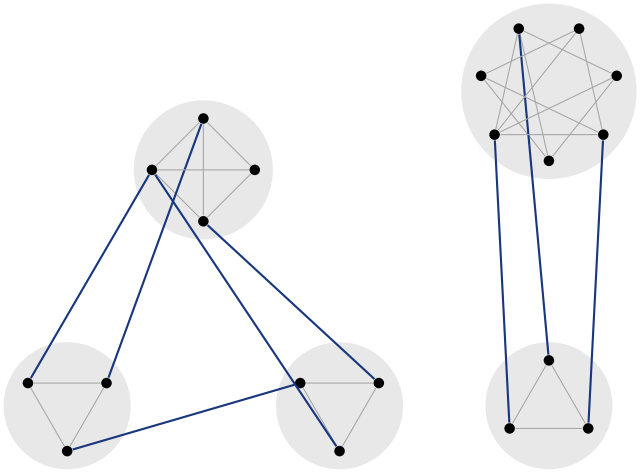


Figure 5: Two extremal graphs for $(10, 3, 3)$, 17 edges. Left: a Bougard–Joret graph, cliques of orders $4, 3, 3$ (shaded) with 5 edges between them. Right: an extremal graph with $\theta(G) = 4$ and $s(G) = 1$, drawn by the two parts of a partition attaining $p(G) = 2$ (Lemma 7): a triangle and the block $X_7$ of Section 8, joined by 3 edges.

(a) Suppose $p \geq 2$. The $q$ edges of $G$ between different sets $V_i$ are not in $H$, and each $V_i$ is a proper nonempty subset, so $|\partial V_i| \geq k$ as in Lemma 1 and $q \geq \lceil kp/2 \rceil$. With Lemma 11, writing $\Delta = t(n + s, \alpha) - t(n, \alpha) - ks \geq 0$,

$$m \;\geq\; \mathrm{e}(H) + q \;\geq\; t(n, \alpha) + (k - 1)s + \Delta + \lceil k(\alpha - s)/2 \rceil \,.$$

Since $\lceil k\alpha/2 \rceil \leq k\alpha/2 + 1/2$ and $\lceil k(\alpha - s)/2 \rceil \geq k(\alpha - s)/2$, this needs $(k/2 - 1)s + \Delta \leq 1/2$. For $k \geq 4$ the left side is at least $s \geq 1$, a contradiction. For $k = 3$ it needs $s = 1$ and then $2 + \Delta + \lceil 3(\alpha - 1)/2 \rceil \leq \lceil 3\alpha/2 \rceil$: for even $\alpha$ the left side is $(3\alpha + 2)/2 + \Delta$, too large; for odd $\alpha$ it is $(3\alpha + 1)/2 + \Delta$, which forces $\Delta = 0$, that is, $\lfloor n/\alpha \rfloor = 3$ by Lemma 11, so $n \leq 4\alpha - 1$.

(b) Suppose $p = 1$, so $s = \alpha - 1$ and $m \geq t(n + \alpha - 1, \alpha) - (\alpha - 1)$. By Lemma 11, $(k - 1)(\alpha - 1) \leq \lceil k\alpha/2 \rceil$, which holds only for $(\alpha, k) \in \{(3, 3), (4, 3), (5, 3), (3, 4), (3, 5)\}$ when $\alpha, k \geq 3$. For each of these pairs the inequality

$$\sum_{i=0}^{\alpha-2} \left\lfloor \frac{n + i}{\alpha} \right\rfloor \;\leq\; \left\lceil \frac{k\alpha}{2} \right\rceil + \alpha - 1, \qquad n > k\alpha, \tag{9}$$

leaves exactly the orders listed: $n \in \{10, 11\}$ for $(\alpha, k) = (3, 3)$, $n = 13$ for $(4, 3)$, $n = 16$ for $(5, 3)$, $n = 13$ for $(3, 4)$ and $n = 16$ for $(3, 5)$. □

**Corollary 13** (conditional on [10, Theorem 3])**.** *For $k \geq 4$ every extremal graph of the second regime splits into $\alpha$ cliques, and so is a Bougard–Joret graph, except possibly at $(13, 3, 4)$ and $(16, 3, 5)$, where $p(G) = 1$. For $k = 3$ the same holds when $\alpha$ is even, except possibly at $(13, 4, 3)$, and when $\alpha$ is odd and $n \geq 4\alpha$, without exception.*

The data match Proposition 12 exactly (Table 4): every observed graph without a partition has $s = 1$ and lies in case (a), or has $s = 2$ at $(10, 3, 3)$, which is case (b). Every extremal graph also satisfies (3) with $s$ computed exactly, none of them with equality, and the prediction that none of the 13 triples analysed last has a graph without a partition was frozen before they were

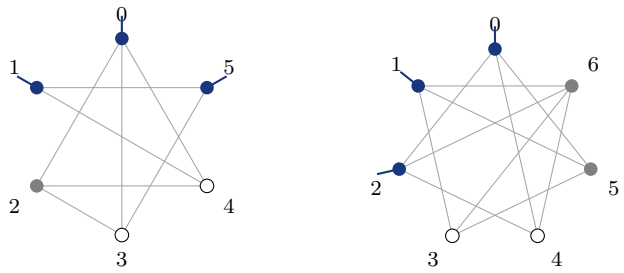



Figure 6: The two building blocks, $X_6$ (left) and $X_7$ (right), each the complement of a maximal triangle-free graph, with independence number 2 and clique cover number 3. Navy vertices with a stub: an attaching triple (for $X_7$, one of three up to automorphism, the others being $\{1,2,3\}$ and $\{1,2,6\}$). Open vertices $3,4$ of $X_6$ and $3,4$ of $X_7$: an independent pair with no neighbour outside the block. Grey vertices: the remaining vertices.

analysed (EXP-011, second addendum). Case (a) is realised at every triple where it is allowed and where we enumerated $E = c$; case (b) is realised only at $(10,3,3)$, by 8 graphs, not at $(11,3,3)$, $(13,4,3)$, $(13,3,4)$, $(16,3,5)$ or $(16,5,3)$. Section 8 shows that case (a) is realised at every triple where it is allowed.

# 8 Extremal graphs without a clique partition for every odd $\alpha$

## 8.1 Two building blocks

The 32 enumerated extremal graphs with $s(G) = 1$ share one shape. Take a partition attaining $p(G) = \alpha - 1$ (Lemma 7). In every one of them, $\alpha - 2$ of its parts are cliques of orders 3 and 4, and the last part $X$ has independence number 2. Moreover $X$ is one of only two graphs:

- $X_6$ (`graph6 ETxO`): six vertices and 8 edges, two more than $K_3 + K_3$;
- $X_7$ (`graph6 FQjRo`): seven vertices and 11 edges, two more than $K_3 + K_4$.

Both are complements of maximal triangle-free graphs. Every part sends exactly three edges to the others, from three distinct vertices, so the edges between parts form a matching, $(3\alpha - 3)/2$ edges in all, and the quotient is a cubic multigraph on the $\alpha - 1$ parts. $X_6$ attaches at one triple of its vertices up to automorphism, and $X_7$ at three (Figure 6).

The count explains why these graphs are extremal. $X$ spends two more edges than the two cliques it replaces, and the quotient on $\alpha - 1$ parts needs $(3\alpha - 3)/2$ edges instead of $\lceil 3\alpha/2 \rceil = (3\alpha + 1)/2$. That is two fewer, so the total is again

$$t(n,\alpha) + 2 + \frac{3\alpha - 3}{2} = t(n,\alpha) + \left\lceil \frac{3\alpha}{2} \right\rceil . \tag{10}$$

The 8 graphs with $s(G) = 2$ at $(10,3,3)$ have a connected $\alpha$-core, the case (b) of Proposition 12. They are 3-connected graphs on ten vertices with independence number 3 and 17 edges, and none of their spanning subgraphs with the same independence number splits into components.

## 8.2 The construction

Let $\alpha \geq 3$ be odd and $3\alpha < n \leq 4\alpha - 1$, and write $n = 3\alpha + r$ with $1 \leq r \leq \alpha - 1$; $T(n,\alpha)$ has $r$ cliques of order 4 and $\alpha - r$ of order 3. Replace two cliques of order 3 by $X_6$ (possible when

$\alpha - r \geq 2$), or a clique of order 3 and one of order 4 by $X_7$ (possible when $r \geq 1$); since $r \geq 1$, at least one replacement is always available, and the data use both. Take a loopless cubic multigraph $Q$ on the resulting $\alpha - 1$ parts that is 3-edge-connected. Such a $Q$ exists because $\alpha - 1$ is even: the triple edge on two parts, $K_4$ on four, and the prism $C_h \square K_2$ on $2h \geq 6$. Map the three edges of $Q$ at each part injectively to vertices of the part: any three vertices of a clique, and an attaching triple $A$ for $X$. Each edge $PR$ of $Q$ becomes an edge of $G$ between the images of its ends. We call these edges *cross edges*; every vertex carries at most one.

**Theorem 14.** *The graph $G$ has $n$ vertices, $t(n, \alpha) + \lceil 3\alpha/2 \rceil$ edges, independence number $\alpha$, connectivity $\kappa(G) \geq 3$ and clique cover number $\theta(G) = \alpha + 1$. In particular $V(G)$ has no partition into $\alpha$ cliques, and $s(G) = 1$.*

The proof uses six facts about the two blocks, each checked exhaustively by a short program (EXP-013, `check_lemmas.py`): for $X \in \{X_6, X_7\}$ and each attaching triple $A$,

(F1) $\alpha(X) = 2$ and $\theta(X) = 3$;

(F2) $X - A$ contains two non-adjacent vertices;

(F3) $\theta(X - S) + |S|/2 \geq 5/2$ for every nonempty $S \subseteq A$;

(F4) $X_6$ has $8 = \mathrm{e}(K_3 + K_3) + 2$ edges and $X_7$ has $11 = \mathrm{e}(K_3 + K_4) + 2$;

(F5) for every set $T$ of at most two vertices, every component of $X - T$ meets $A \setminus T$;

(F6) $X$ is 2-connected.

*Proof. Edges.* The cliques of $G$ span $t(n, \alpha)$ minus the edges of the replaced pair, $X$ spans those edges plus two (F4), and there are $3(\alpha - 1)/2$ cross edges, which gives (10).

*Independence number.* An independent set meets each clique part in at most one vertex and $X$ in at most two (F1), so $\alpha(G) \leq \alpha$. For the converse, F2 gives non-adjacent $x, y \in X \setminus A$; neither carries a cross edge, so neither has a neighbour outside $X$. The cross edges between clique parts form a graph of maximum degree at most one whose vertex classes, the clique parts, have at least $3 \geq 2$ vertices; by Haxell's theorem [18] there is a transversal of the clique parts with no cross edge inside it. With $x$ and $y$ it is an independent set of size $\alpha$.

*Connectivity.* Let $S$ be a set of at most two vertices. Every part keeps a vertex, and every clique part minus $S$ is a nonempty clique. If $X - S$ is connected and $G - S$ were not, the parts would split into two nonempty sides with every edge of $Q$ between them destroyed. But $Q$ has at least three such edges, and each vertex of $S$ destroys at most its own cross edge. If $X - S$ is disconnected, then $S \subseteq V(X)$ by F6, so every other part and every cross edge not at $X$ survives. Moreover $Q - X$ is connected: a cut vertex $v$ of a 3-edge-connected cubic multigraph would leave a side joined to the rest by at most one of the three edges at $v$. So the parts other than $X$ lie in one component. By F5 each component of $X - S$ contains a vertex of $A \setminus S$, whose cross edge leads to a part other than $X$, and so into the single component that holds those parts.

*Clique cover number.* The $\alpha - 2$ cliques and three cliques covering $X$ (F1) give $\theta(G) \leq \alpha + 1$. A clique of $G$ that meets two parts is a single cross edge, since a third vertex would need a second cross edge at one of its ends. Take a partition of $V(G)$ into cliques, let $M'$ be the set of cross edges it uses as cliques, and charge each of them $1/2$ to each of its two parts. A part $P$ then carries

$$c(P) = \theta(P - V(M')) + \tfrac{1}{2}\,|P \cap V(M')|.$$

A clique part carries at least 1: either some vertex is left, needing one more clique, or all of its at least three vertices are covered, giving at least 3/2. For $X$, the covered vertices form a set $S \subseteq A$, and $c(X) = \theta(X) = 3$ when $S$ is empty and $c(X) \geq 5/2$ otherwise, by F3. The partition has $\sum_P c(P) \geq \alpha - 2 + 5/2$ cliques, so at least $\alpha + 1$.

*Core parameter.* The parts of $G$ have independence numbers summing to $\alpha$, so $p(G) \geq \alpha - 1$ by Lemma 7, and $p(G) \neq \alpha$ because $\theta(G) > \alpha$. □

**Corollary 15.** *At each of the nine triples with $n \leq 24$ that it covers, $(10, 3, 3)$, $(11, 3, 3)$, $(16, 5, 3)$ to $(19, 5, 3)$ and $(22, 7, 3)$ to $(24, 7, 3)$, the graph $G$ is extremal by Theorem 8, without any condition, and has no partition into $\alpha$ cliques. If Didin's Theorem 9 holds, $G$ is extremal for every odd $\alpha \geq 3$ and $3\alpha < n \leq 4\alpha - 1$; then case (a) of Proposition 12 occurs at every triple where it is allowed, and the bound $n \leq 4\alpha - 1$ there is sharp.*

A program that builds $G$ with the triple edge ($\alpha = 3$), $K_4$ ($\alpha = 5$) or the prism $C_h \square K_2$ ($\alpha \geq 7$) as quotient and checks every property of Theorem 14 exactly confirms it for every odd $\alpha \leq 11$ and every admissible order (EXP-013, 30 graphs). It also confirms that a quotient which is not 3-edge-connected can fail: the circular ladder on four parts, two double edges joined by two single edges, has a cut of two edges, and all 80 graphs built on it at $\alpha = 5$, twenty random attachments at each order from 16 to 19, have the conjectured number of edges and independence number 5 but connectivity 2 (EXP-013, `probe_ladder.py`).

## 9 Discussion

**Relation to Didin's proof.** Our computation is independent of Didin's argument: it does not use inequality (3) or its proof. On every open-window triple with $n \leq 24$ $f$ equals the bound of Didin's Theorem 9. It cannot prove the general statement, and its agreement is evidence about a finite range only. Conversely, Proposition 12 uses Didin's main theorem and shows that his inequality, if correct, already confines the non-Bougard–Joret extremal graphs to a thin family; the data neither contradict the inequality nor reach its equality case on these graphs.

**Why a proof by clique partition cannot work at equality.** Proposition 8 of Bougard and Joret and Proposition 4.2 of Das and Gupta reduce the second line to the existence of a partition into $\alpha$ cliques below the conjectured size. The graphs with $\theta(G) = \alpha + 1$ show that the reduction cannot be pushed to the conjectured size itself: at $k = 3$ and odd $\alpha$ there are extremal graphs without such a partition. A proof must therefore bound the size strictly below $c$ only, as Lemma 1 does, or proceed without the partition, as Didin's does.

The 8 graphs with $s(G) = 2$ at $(10, 3, 3)$ form the only case (b) found. Their clique number is 3, their degrees are 3 to 5, and their automorphism groups have order at most 4; none of them splits into two parts whose independence numbers sum to 3, so each has a connected $\alpha$-core. None is tight in Didin's sense: each has strictly more than $t(n + s, \alpha) - s = t(12, 3) - 2 = 16$ edges. This agrees with his Lemma 18, which shows that the tight graphs with $(b, \alpha, n) = (3, 3, 10)$ have a 2-vertex cut and so are not 3-connected.

**The graphs attaining the second line.** Didin's preprint characterises the graphs attaining his main inequality, but not those attaining Theorem 9. In the range enumerated here, the extremal graphs are of three kinds: the Bougard–Joret graphs; the graphs of Theorem 14, which account for every extremal graph with $s(G) = 1$ (a 2-edge cut of the quotient would contradict 3-connectivity);

and 8 graphs at $(10, 3, 3)$ with a connected $\alpha$-core. Whether this list is complete for every order is a natural conjecture. Given Didin's theorem, Lemma 10 and Proposition 12 prove part of it outside case (b). An extremal graph without a partition into $\alpha$ cliques has $k = 3$, $\alpha$ odd, $s(G) = 1$ and $n \leq 4\alpha - 1$, and equality throughout the proof of Proposition 12(a) forces three more facts: the $\alpha$-core $H$ attains (3), so it is tight in Didin's sense; exactly $(3\alpha - 3)/2$ edges join its $\alpha - 1$ components; and every component sends exactly three of them. The proof does not show that the components with independence number 1 are cliques of orders 3 and 4, that the edges between components form a matching, that the component with independence number 2 is $X_6$ or $X_7$, or that it attaches only at the observed triples. These four claims hold for every enumerated graph and are open in general.

**Open questions.** Theorem 14 realises case (a) of Proposition 12 everywhere. Two questions remain. First, whether the four structural claims above hold for every odd $\alpha$, in particular whether $X_6$ and $X_7$ are the only blocks, which the enumeration shows for $n \leq 22$ only. Second, whether case (b) is realised beyond $(10, 3, 3)$, which the enumeration rules out at the other five triples it allows. The value of $f$ beyond order 24 is decided by Didin's theorem if it is correct. For the enumeration, the factors of Section 6 between consecutive orders, if they persisted, would put order 25 at several days of the machine used here; this is an extrapolation, not a measurement.

**What the validation does not cover.** The gates sample the production streams; they do not re-run them. Gate B covers 21,196 graphs in slices, not the billions of the full search, and 19 of its 33 slices are empty, including every slice at order 24 and every production edge count with $k = 7$. At those parameters the filter is covered by Lemmas 4 and 5 and by Gate A, on constructed graphs up to order 24 with $k' \leq 9$, not by production streams. Gate E checks the generator's counts for connected, cubic and quartic graphs up to order 22 with known values; no published count exists for the restricted classes the search enumerates (minimum degree $k$, independence number at most $\alpha$, a fixed number of edges), so the completeness of those streams rests on the generator's correctness and on the soundness of the hook (Lemma 3), which Gate C tests directly.

**Limitations.** The values rest on the correctness of `geng` and of the split into parts, which we test but do not prove (Gates D and E); on the filter, which we prove (Lemmas 4 and 5) and test (Gates A and B); and on the hook, which we prove (Lemma 3) and test (Gate C). The binaries are built for the processor they ran on (`-march=native`); a rebuild elsewhere gives other bytes with the same source. Didin's results are used as stated, without an audit of their proofs.

## 10 Conclusions

The second line of the Bougard–Joret conjecture holds on every open-window triple with $n \leq 24$: 53 triples, decided by complete enumeration of 6,223,452,780 candidate graphs and an exact filter, with no triple left undecided and no edge count left capped. For seven pairs $(\alpha, k)$ the open window ends by order 24, so the conjectured value holds there for every order, independently of Didin's argument. The search is a finite decision procedure with five components. Three of them, the prune, the independence test and the connectivity test, are proved correct and also tested against an independent implementation. The other two, the generator and its split into parts, are tested and not proved: the generator against known counts up to order 22, and the split at three triples of orders 17 to 20. The binaries that ran are reproduced byte for byte from the committed sources.

Table 5: The experiments behind this paper, with the commit that froze each hypothesis before any computation (the hypothesis of EXP-011 carries two addenda, each frozen before the analysis it predicts).

| experiment | frozen at | content |
|---|---|---|
| EXP-005 | `ad34cd90` | the seven smallest open triples, plain generator |
| EXP-006 | `c04a8a8f` | the hook and the early-stopping filter; four triples |
| EXP-007 | `f85ce6b9` | every open triple up to order 22 |
| EXP-008 | `19125234` | order 23 |
| EXP-009 | `153f1166` | order 24, four of six triples |
| EXP-010 | `084e0874` | the two remaining order-24 edge counts |
| EXP-011 | `47c86e8f`, `a480c533`, `e552f1c6` | every extremal graph at 35 triples; structure; $p(G)$ |
| EXP-012 | `78cc42fb` | validation at the production parameters |
| EXP-013 | `15a64e33` | the construction for odd $\alpha$ |

The values carry no statistical uncertainty. Every number in Theorem 8 and in Tables 3 and 4 is an exact count from a complete enumeration or an exact test, and the run-to-run variation of the wall-clock times does not enter any value. The residual risk is of a different kind: a defect in `geng` or in its split into parts would lose graphs silently. Gate E tests the generator on the classes with published counts, up to order 22, and Gate D tests the split at orders 17 to 20 with the hook active; neither reaches orders 23 and 24 on the restricted classes the search enumerates, and Section 9 lists what the gates do not cover.

At the conjectured value the extremal graphs are not all of Bougard and Joret's form. At 35 triples, 40 of the 246 extremal graphs have no partition into $\alpha$ cliques; all of them have $k = 3$ and $\alpha$ odd. Didin's inequality, if correct, confines such graphs to $n \leq 4\alpha - 1$ or six small triples, and the data agree with that prediction, which was frozen before the last 13 triples were analysed. The two blocks $X_6$ and $X_7$ found in the data give a construction for every odd $\alpha$ and every $3\alpha < n \leq 4\alpha - 1$, proved with six finite facts checked exactly. A proof of the second line through a partition into $\alpha$ cliques therefore cannot reach the conjectured size itself, and the structure of the extremal graphs beyond the enumerated range rests on the four open claims of Section 9 and on Didin's theorem.

## 11 Data, code and reproducibility

All code and records are in the repository `github.com/fsantibanezleal/CAOS_RESEARCH`, under `problems/combinatorics/bougard-joret/ experiments/`, experiments EXP-005 to EXP-013. Each experiment holds its question and predictions, fixed in a commit before its computation, its scripts, the raw records (`artifacts/enum.jsonl`: order, $\alpha$, $k$, edge count, graphs generated, graphs kept, up to five kept examples, number of parts, seconds) and a written statement of its result. The extremal graphs are in `EXP-011-extremal-structure/artifacts/extremal/`, one `graph6` file per triple, with the per-graph analysis in `structure.jsonl`, `verify.jsonl` and `didin.jsonl`.

**Experiments.** Table 5 lists, for each experiment, the commit that fixed its question and predictions before the computation.

A failed or partial outcome stays in the record with its experiment: the first-pass caps of EXP-009, the harness defects of the first Gate D and Gate E runs in EXP-012, and the refuted structural prediction of EXP-011 (that every extremal graph splits into $\alpha$ cliques).

**Reproducing one value.** The value $f(16, 5, 3) = 26$ is a short example. Here $t(16, 5) = 18$, $c = 26$ and $\ell = \max(24, 18) = 24$, so two edge counts are searched. With the pruned generator, $E = 24$ yields 13 candidates and $E = 25$ yields 531, none of them qualifying, in under a second; the plain generator yields 4,207 and 927,968 candidates at the same counts, again none qualifying. At $E = c = 26$ the pruned generator yields 18,979 candidates, of which exactly 18 qualify; these are the extremal graphs of Table 4, 13 of them Bougard–Joret graphs and 5 with $s(G) = 1$.

**Binaries.** SHA-256 of the binaries that ran, each reproduced byte for byte by Gate F:

```
filter2 ebbfd34da19290f5f08b0c1b5e2441eab89a7a3d36894933be2f2e490269b02f
geng_a3 43259cfa498b57af9bacea605cd7b7517361d908b4650d6f55631b3b0c4b53c9
geng_a4 124d5588886dcbfd4fa837573bc02e65a2a7ccbb3497fdface3855e233aea9fa
geng_a5 d3bf334991bf37ffb5428a5bc43b6ad2191e70f4c87e52a136b8787e77e61419
geng_a6 69e35a9660ce409ab2fbbe043aaca82196580e284dda13aa64d3dd64846b290c
geng_a7 b98aac98b1645dadf0bfb6fe68aab2b467f83a264ed8751316a3bde84a9413f9
```

**Software.** nauty 2.8.9 (`geng.c` 3.7), gcc 13.3.0, Ubuntu 24.04 under WSL2, Python 3.12 in WSL and Python 3.13 with NetworkX 3.4.1 on Windows. Build lines:

```
gcc -O2 -o filter2 filter2.c
gcc -O4 -mpopcnt -march=native -o geng_aA -DMAXN=WORDSIZE -DWORDSIZE=32
-DPRUNE=prune_alpha -DALPHA_BOUND=A geng.c prune_alpha.c
gtoolsW.o nautyW1.o nautilW1.o naugraphW1.o schreier.o naurng.o
```

**Commands.** One edge count of one triple, for example $(22, 5, 3)$ at $E = 45$:

```
python3 run_enum.py <nauty>/geng_a5 <bin>/filter2 22 5 3 45 45
--jobs 28 --mod 56
```

The extremal graphs and their analysis: `collect.py`, then `analyze.py`, `verify.py` and `didin.py` in EXP-011; the validation: `build.sh` and `validate.py` in EXP-012. The tables and figures of this paper are generated from the records by `make_tables.py` in the manuscript folder.

**Evidence marks.** Values and counts are machine-verified exhaustive computations; the lemmas, propositions and theorems are proved here; Proposition 12 and Corollary 13 are conditional on Didin's Theorem 3; Didin's statements were read on the primary source and not audited.

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# A Every enumeration record

Table 6 lists every record of EXP-005 to EXP-010, one row per triple, edge count and run, with the generator, the number of parts, the graphs generated and kept, and the wall-clock seconds. A triple appears more than once when an edge count was run with both generators or rerun after a cap. The totals of Section 6 are sums over these rows. The rows are ordered by triple and edge count, and each one is a line of the file `artifacts/enum.jsonl` of the experiment named in its record column, written by that experiment's `run_enum.py` when the edge count finished. A row below the conjectured value with no kept graph is the evidence for one step of Algorithm 3. A row at $E = c$, where an experiment ran that count, gives the number of extremal graphs, and the regeneration of Section 7 reproduces it (the column "recorded" of Table 4). Seconds are wall-clock times of the whole edge count over all its parts, so rows run on different numbers of processes are not comparable as CPU time.

Table 6: Every enumeration record of EXP-005 to EXP-010. Generator: *plain* is `geng` with minimum degree $k$; *pruned* adds the hook of Lemma 3. Parts: the number $M$ of disjoint parts. Kept: graphs with independence number $\alpha$ and connectivity at least $k$. Seconds: wall-clock time.

| triple | $E$ | record | generator | parts | graphs | kept | seconds |
|---|---|---|---|---|---|---|---|
| $(10,3,3)$ | 15 | EXP-005 | plain | 48 | 21 | 0 | 0.1 |
| $(10,3,3)$ | 16 | EXP-005 | plain | 48 | 362 | 0 | 0.1 |
| $(10,3,3)$ | 17 | EXP-005 | plain | 48 | 3,484 | 15 | 0.1 |
| $(11,3,3)$ | 17 | EXP-005 | plain | 48 | 214 | 0 | 0.1 |
| $(11,3,3)$ | 18 | EXP-005 | plain | 48 | 4,982 | 0 | 0.1 |
| $(11,3,3)$ | 19 | EXP-005 | plain | 48 | 50,616 | 0 | 0.1 |
| $(11,3,3)$ | 20 | EXP-005 | plain | 48 | 302,901 | 6 | 0.1 |
| $(12,3,3)$ | 18 | EXP-005 | plain | 96 | 94 | 0 | 0.2 |
| $(12,3,3)$ | 19 | EXP-005 | plain | 96 | 4,338 | 0 | 0.2 |
| $(12,3,3)$ | 20 | EXP-005 | plain | 96 | 85,599 | 0 | 0.2 |
| $(12,3,3)$ | 21 | EXP-005 | plain | 96 | 878,510 | 0 | 0.3 |
| $(12,3,3)$ | 22 | EXP-005 | plain | 96 | 5,738,052 | 0 | 0.7 |
| $(12,3,3)$ | 23 | EXP-005 | plain | 96 | 26,837,832 | 2 | 2.1 |
| $(13,3,4)$ | 26 | EXP-005 | plain | 224 | 10,786 | 0 | 0.4 |
| $(13,3,4)$ | 27 | EXP-005 | plain | 224 | 1,084,246 | 0 | 1.6 |
| $(13,3,4)$ | 28 | EXP-005 | plain | 224 | 24,242,521 | 1 | 6.0 |
| $(13,4,3)$ | 20 | EXP-005 | plain | 224 | 1,999 | 0 | 0.3 |
| $(13,4,3)$ | 21 | EXP-005 | plain | 224 | 90,446 | 1 | 0.4 |

| triple | $E$ | record | generator | parts | graphs | kept | seconds |
|---|---|---|---|---|---|---|---|
| $(14,3,4)$ | 28 | EXP-005 | plain | 896 | 88,193 | 0 | 2.0 |
| $(14,3,4)$ | 29 | EXP-005 | plain | 896 | 12,303,616 | 0 | 20.9 |
| $(14,3,4)$ | 30 | EXP-005 | plain | 896 | 358,456,307 | 0 | 111.5 |
| $(14,3,4)$ | 31 | EXP-005 | plain | 896 | 4,898,113,345 | 0 | 620.0 |
| $(14,4,3)$ | 21 | EXP-005 | plain | 448 | 540 | 0 | 0.8 |
| $(14,4,3)$ | 22 | EXP-005 | plain | 448 | 59,827 | 0 | 0.9 |
| $(14,4,3)$ | 23 | EXP-005 | plain | 448 | 2,086,675 | 0 | 2.2 |
| $(14,4,3)$ | 24 | EXP-005 | plain | 448 | 35,850,020 | 1 | 10.3 |
| $(15,4,3)$ | 23 | EXP-005 | plain | 896 | 22,186 | 0 | 2.6 |
| $(15,4,3)$ | 24 | EXP-005 | plain | 896 | 1,736,751 | 0 | 11.3 |
| $(15,4,3)$ | 25 | EXP-005 | plain | 896 | 51,457,768 | 0 | 49.0 |
| $(15,4,3)$ | 26 | EXP-005 | plain | 896 | 850,994,936 | 0 | 220.7 |
| $(15,4,3)$ | 27 | EXP-005 | plain | 896 | 9,384,504,479 | 1 | 1210.4 |
| $(16,3,5)$ | 40 | EXP-006 | pruned | 56 | 2 | 0 | 0.1 |
| $(16,3,5)$ | 41 | EXP-006 | pruned | 56 | 10 | 0 | 0.1 |
| $(16,3,5)$ | 42 | EXP-006 | pruned | 56 | 122 | 0 | 0.3 |
| $(16,3,5)$ | 43 | EXP-006 | pruned | 56 | 1,344 | 3 | 0.8 |
| $(16,5,3)$ | 24 | EXP-006 | plain | 448 | 4,207 | 0 | 0.7 |
| $(16,5,3)$ | 24 | EXP-006 | plain | 896 | 4,207 | 0 | 1.5 |
| $(16,5,3)$ | 24 | EXP-006 | pruned | 56 | 13 | 0 | 0.1 |
| $(16,5,3)$ | 25 | EXP-006 | plain | 896 | 927,968 | 0 | 33.0 |
| $(16,5,3)$ | 25 | EXP-006 | pruned | 56 | 531 | 0 | 0.4 |
| $(16,5,3)$ | 26 | EXP-006 | pruned | 56 | 18,979 | 18 | 1.3 |
| $(17,3,5)$ | 43 | EXP-007 | pruned | 56 | 0 | 0 | 0.2 |
| $(17,3,5)$ | 44 | EXP-007 | pruned | 56 | 2 | 0 | 0.3 |
| $(17,3,5)$ | 45 | EXP-007 | pruned | 56 | 32 | 0 | 0.5 |
| $(17,3,5)$ | 46 | EXP-007 | pruned | 56 | 254 | 0 | 0.9 |
| $(17,3,5)$ | 47 | EXP-007 | pruned | 56 | 1,832 | 0 | 2.3 |
| $(17,3,5)$ | 48 | EXP-007 | pruned | 56 | 12,267 | 3 | 4.4 |
| $(17,4,4)$ | 34 | EXP-006 | pruned | 56 | 3 | 0 | 0.2 |
| $(17,4,4)$ | 35 | EXP-006 | pruned | 56 | 96 | 0 | 1.0 |
| $(17,4,4)$ | 36 | EXP-006 | pruned | 56 | 3,000 | 2 | 3.7 |
| $(17,5,3)$ | 26 | EXP-006 | plain | 1792 | 287,045 | 0 | 95.6 |
| $(17,5,3)$ | 26 | EXP-006 | pruned | 56 | 16 | 0 | 0.1 |
| $(17,5,3)$ | 27 | EXP-006 | plain | 1792 | 35,329,226 | 0 | 1040.2 |
| $(17,5,3)$ | 27 | EXP-006 | pruned | 56 | 482 | 0 | 0.4 |
| $(17,5,3)$ | 28 | EXP-006 | pruned | 56 | 11,282 | 0 | 1.4 |
| $(17,5,3)$ | 29 | EXP-006 | pruned | 56 | 218,377 | 34 | 7.3 |
| $(18,3,5)$ | 45 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(18,3,5)$ | 46 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(18,3,5)$ | 47 | EXP-007 | pruned | 56 | 4 | 0 | 0.3 |
| $(18,3,5)$ | 48 | EXP-007 | pruned | 56 | 14 | 0 | 0.4 |
| $(18,3,5)$ | 49 | EXP-007 | pruned | 56 | 56 | 0 | 0.4 |
| $(18,3,5)$ | 50 | EXP-007 | pruned | 56 | 231 | 0 | 0.9 |
| $(18,3,5)$ | 51 | EXP-007 | pruned | 56 | 1,089 | 0 | 1.9 |
| $(18,3,5)$ | 52 | EXP-007 | pruned | 56 | 5,172 | 0 | 4.7 |
| $(18,3,5)$ | 53 | EXP-007 | pruned | 56 | 25,882 | 2 | 10.7 |
| $(18,4,4)$ | 36 | EXP-007 | pruned | 56 | 2 | 0 | 0.2 |
| $(18,4,4)$ | 37 | EXP-007 | pruned | 56 | 18 | 0 | 0.6 |
| $(18,4,4)$ | 38 | EXP-007 | pruned | 56 | 263 | 0 | 2.0 |
| $(18,4,4)$ | 39 | EXP-007 | pruned | 56 | 3,803 | 0 | 6.0 |
| $(18,4,4)$ | 40 | EXP-007 | pruned | 56 | 50,279 | 4 | 30.1 |
| $(18,5,3)$ | 27 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(18,5,3)$ | 28 | EXP-007 | pruned | 56 | 20 | 0 | 0.2 |
| $(18,5,3)$ | 29 | EXP-007 | pruned | 56 | 306 | 0 | 0.5 |
| $(18,5,3)$ | 30 | EXP-007 | pruned | 56 | 4,683 | 0 | 1.3 |
| $(18,5,3)$ | 31 | EXP-007 | pruned | 56 | 66,011 | 0 | 7.6 |
| $(18,5,3)$ | 32 | EXP-007 | pruned | 56 | 876,548 | 35 | 15.4 |
| $(19,3,5)$ | 51 | EXP-007 | pruned | 56 | 1 | 0 | 0.7 |
| $(19,3,5)$ | 52 | EXP-007 | pruned | 56 | 2 | 0 | 0.7 |
| $(19,3,5)$ | 53 | EXP-007 | pruned | 56 | 9 | 0 | 0.7 |
| $(19,3,5)$ | 54 | EXP-007 | pruned | 56 | 34 | 0 | 0.7 |
| $(19,3,5)$ | 55 | EXP-007 | pruned | 56 | 150 | 0 | 1.0 |
| $(19,3,5)$ | 56 | EXP-007 | pruned | 56 | 646 | 0 | 2.5 |
| $(19,3,5)$ | 57 | EXP-007 | pruned | 56 | 2,997 | 0 | 5.0 |
| $(19,3,5)$ | 58 | EXP-007 | pruned | 56 | 14,081 | 0 | 15.0 |

| triple | $E$ | record | generator | parts | graphs | kept | seconds |
|---|---|---|---|---|---|---|---|
| $(19,3,5)$ | 59 | EXP-007 | pruned | 56 | 68,442 | 4 | 27.3 |
| $(19,3,6)$ | 57 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(19,3,6)$ | 58 | EXP-007 | pruned | 56 | 7 | 0 | 2.5 |
| $(19,3,6)$ | 59 | EXP-007 | pruned | 56 | 75 | 0 | 6.6 |
| $(19,3,6)$ | 60 | EXP-007 | pruned | 56 | 880 | 1 | 23.2 |
| $(19,4,4)$ | 38 | EXP-007 | pruned | 56 | 0 | 0 | 0.1 |
| $(19,4,4)$ | 39 | EXP-007 | pruned | 56 | 2 | 0 | 0.2 |
| $(19,4,4)$ | 40 | EXP-007 | pruned | 56 | 25 | 0 | 0.4 |
| $(19,4,4)$ | 41 | EXP-007 | pruned | 56 | 242 | 0 | 1.0 |
| $(19,4,4)$ | 42 | EXP-007 | pruned | 56 | 2,353 | 0 | 4.4 |
| $(19,4,4)$ | 43 | EXP-007 | pruned | 56 | 22,051 | 0 | 12.9 |
| $(19,4,4)$ | 44 | EXP-007 | pruned | 56 | 201,009 | 2 | 33.7 |
| $(19,5,3)$ | 29 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(19,5,3)$ | 30 | EXP-007 | pruned | 56 | 12 | 0 | 0.2 |
| $(19,5,3)$ | 31 | EXP-007 | pruned | 56 | 110 | 0 | 0.3 |
| $(19,5,3)$ | 32 | EXP-007 | pruned | 56 | 1,151 | 0 | 0.5 |
| $(19,5,3)$ | 33 | EXP-007 | pruned | 56 | 12,420 | 0 | 1.5 |
| $(19,5,3)$ | 34 | EXP-007 | pruned | 56 | 135,903 | 0 | 5.9 |
| $(19,5,3)$ | 35 | EXP-007 | pruned | 56 | 1,478,543 | 18 | 27.5 |
| $(19,6,3)$ | 29 | EXP-007 | pruned | 56 | 295 | 0 | 2.1 |
| $(19,6,3)$ | 30 | EXP-007 | pruned | 56 | 24,882 | 2 | 32.6 |
| $(20,3,5)$ | 57 | EXP-007 | pruned | 56 | 1 | 0 | 1.1 |
| $(20,3,5)$ | 58 | EXP-007 | pruned | 56 | 3 | 0 | 0.9 |
| $(20,3,5)$ | 59 | EXP-007 | pruned | 56 | 11 | 0 | 0.9 |
| $(20,3,5)$ | 60 | EXP-007 | pruned | 56 | 43 | 0 | 1.1 |
| $(20,3,5)$ | 61 | EXP-007 | pruned | 56 | 184 | 0 | 2.2 |
| $(20,3,5)$ | 62 | EXP-007 | pruned | 56 | 792 | 0 | 4.5 |
| $(20,3,5)$ | 63 | EXP-007 | pruned | 56 | 3,597 | 0 | 11.6 |
| $(20,3,5)$ | 64 | EXP-007 | pruned | 56 | 16,741 | 0 | 29.6 |
| $(20,3,5)$ | 65 | EXP-007 | pruned | 56 | 80,288 | 4 | 47.7 |
| $(20,3,6)$ | 60 | EXP-007 | pruned | 56 | 0 | 0 | 0.3 |
| $(20,3,6)$ | 61 | EXP-007 | pruned | 56 | 0 | 0 | 1.6 |
| $(20,3,6)$ | 62 | EXP-007 | pruned | 56 | 3 | 0 | 4.3 |
| $(20,3,6)$ | 63 | EXP-007 | pruned | 56 | 52 | 0 | 8.5 |
| $(20,3,6)$ | 64 | EXP-007 | pruned | 56 | 443 | 0 | 24.6 |
| $(20,3,6)$ | 65 | EXP-007 | pruned | 56 | 3,548 | 0 | 64.2 |
| $(20,3,6)$ | 66 | EXP-007 | pruned | 56 | 26,008 | 1 | 209.0 |
| $(20,4,4)$ | 40 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(20,4,4)$ | 41 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(20,4,4)$ | 42 | EXP-007 | pruned | 56 | 5 | 0 | 0.4 |
| $(20,4,4)$ | 43 | EXP-007 | pruned | 56 | 22 | 0 | 0.8 |
| $(20,4,4)$ | 44 | EXP-007 | pruned | 56 | 121 | 0 | 1.0 |
| $(20,4,4)$ | 45 | EXP-007 | pruned | 56 | 735 | 0 | 1.5 |
| $(20,4,4)$ | 46 | EXP-007 | pruned | 56 | 5,020 | 0 | 3.6 |
| $(20,4,4)$ | 47 | EXP-007 | pruned | 56 | 35,903 | 0 | 14.8 |
| $(20,4,4)$ | 48 | EXP-007 | pruned | 56 | 266,577 | 2 | 53.2 |
| $(20,5,3)$ | 30 | EXP-007 | pruned | 56 | 1 | 0 | 0.1 |
| $(20,5,3)$ | 31 | EXP-007 | pruned | 56 | 1 | 0 | 0.2 |
| $(20,5,3)$ | 32 | EXP-007 | pruned | 56 | 5 | 0 | 0.2 |
| $(20,5,3)$ | 33 | EXP-007 | pruned | 56 | 27 | 0 | 0.6 |
| $(20,5,3)$ | 34 | EXP-007 | pruned | 56 | 188 | 0 | 0.9 |
| $(20,5,3)$ | 35 | EXP-007 | pruned | 56 | 1,429 | 0 | 1.1 |
| $(20,5,3)$ | 36 | EXP-007 | pruned | 56 | 12,641 | 0 | 2.5 |
| $(20,5,3)$ | 37 | EXP-007 | pruned | 56 | 118,362 | 0 | 12.8 |
| $(20,5,3)$ | 38 | EXP-007 | pruned | 56 | 1,147,270 | 5 | 41.2 |
| $(20,6,3)$ | 30 | EXP-007 | pruned | 56 | 13 | 0 | 0.2 |
| $(20,6,3)$ | 31 | EXP-007 | pruned | 56 | 556 | 0 | 4.3 |
| $(20,6,3)$ | 32 | EXP-007 | pruned | 56 | 22,747 | 0 | 24.4 |
| $(20,6,3)$ | 33 | EXP-007 | pruned | 56 | 720,208 | 5 | 164.8 |
| $(21,3,6)$ | 63 | EXP-007 | pruned | 56 | 1 | 0 | 0.4 |
| $(21,3,6)$ | 64 | EXP-007 | pruned | 56 | 1 | 0 | 1.3 |
| $(21,3,6)$ | 65 | EXP-007 | pruned | 56 | 4 | 0 | 2.5 |
| $(21,3,6)$ | 66 | EXP-007 | pruned | 56 | 14 | 0 | 3.5 |
| $(21,3,6)$ | 67 | EXP-007 | pruned | 56 | 56 | 0 | 7.0 |
| $(21,3,6)$ | 68 | EXP-007 | pruned | 56 | 225 | 0 | 12.1 |
| $(21,3,6)$ | 69 | EXP-007 | pruned | 56 | 1,019 | 0 | 26.1 |

| triple | $E$ | record | generator | parts | graphs | kept | seconds |
|---|---|---|---|---|---|---|---|
| $(21,3,6)$ | 70 | EXP-007 | pruned | 56 | 4,726 | 0 | 75.2 |
| $(21,3,6)$ | 71 | EXP-007 | pruned | 56 | 23,045 | 0 | 213.4 |
| $(21,3,6)$ | 72 | EXP-007 | pruned | 56 | 116,011 | 1 | 465.3 |
| $(21,4,4)$ | 45 | EXP-007 | pruned | 56 | 1 | 0 | 1.6 |
| $(21,4,4)$ | 46 | EXP-007 | pruned | 56 | 2 | 0 | 4.6 |
| $(21,4,4)$ | 47 | EXP-007 | pruned | 56 | 12 | 0 | 2.4 |
| $(21,4,4)$ | 48 | EXP-007 | pruned | 56 | 64 | 0 | 2.3 |
| $(21,4,4)$ | 49 | EXP-007 | pruned | 56 | 396 | 0 | 3.5 |
| $(21,4,4)$ | 50 | EXP-007 | pruned | 56 | 2,542 | 0 | 7.0 |
| $(21,4,4)$ | 51 | EXP-007 | pruned | 56 | 17,349 | 0 | 19.8 |
| $(21,4,4)$ | 52 | EXP-007 | pruned | 56 | 121,484 | 0 | 71.4 |
| $(21,4,4)$ | 53 | EXP-007 | pruned | 56 | 872,388 | 2 | 247.2 |
| $(21,4,5)$ | 53 | EXP-007 | pruned | 56 | 10 | 0 | 20.9 |
| $(21,4,5)$ | 54 | EXP-007 | pruned | 56 | 410 | 0 | 211.3 |
| $(21,4,5)$ | 55 | EXP-007 | pruned | 56 | 12,504 | 1 | 2642.7 |
| $(21,5,3)$ | 34 | EXP-007 | pruned | 56 | 1 | 0 | 0.6 |
| $(21,5,3)$ | 35 | EXP-007 | pruned | 56 | 2 | 0 | 1.3 |
| $(21,5,3)$ | 36 | EXP-007 | pruned | 56 | 13 | 0 | 1.7 |
| $(21,5,3)$ | 37 | EXP-007 | pruned | 56 | 83 | 0 | 1.9 |
| $(21,5,3)$ | 38 | EXP-007 | pruned | 56 | 631 | 0 | 1.6 |
| $(21,5,3)$ | 39 | EXP-007 | pruned | 56 | 5,151 | 0 | 3.1 |
| $(21,5,3)$ | 40 | EXP-007 | pruned | 56 | 45,680 | 0 | 8.1 |
| $(21,5,3)$ | 41 | EXP-007 | pruned | 56 | 420,350 | 0 | 86.7 |
| $(21,5,3)$ | 42 | EXP-007 | pruned | 56 | 3,958,285 | 17 | 221.6 |
| $(21,5,4)$ | 42 | EXP-007 | pruned | 56 | 24 | 0 | 4.0 |
| $(21,5,4)$ | 43 | EXP-007 | pruned | 56 | 1,274 | 0 | 200.4 |
| $(21,5,4)$ | 44 | EXP-007 | pruned | 56 | 56,482 | 7 | 1186.3 |
| $(21,6,3)$ | 32 | EXP-007 | pruned | 56 | 16 | 0 | 0.3 |
| $(21,6,3)$ | 33 | EXP-007 | pruned | 56 | 494 | 0 | 2.5 |
| $(21,6,3)$ | 34 | EXP-007 | pruned | 56 | 12,663 | 0 | 13.7 |
| $(21,6,3)$ | 35 | EXP-007 | pruned | 56 | 290,442 | 0 | 76.2 |
| $(21,6,3)$ | 36 | EXP-007 | pruned | 56 | 5,968,993 | 5 | 526.0 |
| $(22,3,6)$ | 70 | EXP-007 | pruned | 56 | 1 | 0 | 4.8 |
| $(22,3,6)$ | 71 | EXP-007 | pruned | 56 | 2 | 0 | 7.1 |
| $(22,3,6)$ | 72 | EXP-007 | pruned | 56 | 9 | 0 | 4.7 |
| $(22,3,6)$ | 73 | EXP-007 | pruned | 56 | 34 | 0 | 4.9 |
| $(22,3,6)$ | 74 | EXP-007 | pruned | 56 | 150 | 0 | 8.5 |
| $(22,3,6)$ | 75 | EXP-007 | pruned | 56 | 639 | 0 | 20.3 |
| $(22,3,6)$ | 76 | EXP-007 | pruned | 56 | 2,935 | 0 | 43.2 |
| $(22,3,6)$ | 77 | EXP-007 | pruned | 56 | 13,706 | 0 | 111.8 |
| $(22,3,6)$ | 78 | EXP-007 | pruned | 56 | 66,455 | 0 | 427.8 |
| $(22,3,6)$ | 79 | EXP-007 | pruned | 56 | 330,190 | 1 | 1300.9 |
| $(22,3,7)$ | 77 | EXP-007 | pruned | 56 | 1 | 0 | 2.3 |
| $(22,3,7)$ | 78 | EXP-007 | pruned | 56 | 6 | 0 | 64.0 |
| $(22,3,7)$ | 79 | EXP-007 | pruned | 56 | 66 | 0 | 318.8 |
| $(22,3,7)$ | 80 | EXP-007 | pruned | 56 | 709 | 0 | 1331.4 |
| $(22,4,4)$ | 50 | EXP-007 | pruned | 56 | 1 | 0 | 4.1 |
| $(22,4,4)$ | 51 | EXP-007 | pruned | 56 | 4 | 0 | 4.8 |
| $(22,4,4)$ | 52 | EXP-007 | pruned | 56 | 19 | 0 | 5.0 |
| $(22,4,4)$ | 53 | EXP-007 | pruned | 56 | 103 | 0 | 5.5 |
| $(22,4,4)$ | 54 | EXP-007 | pruned | 56 | 638 | 0 | 10.7 |
| $(22,4,4)$ | 55 | EXP-007 | pruned | 56 | 4,077 | 0 | 15.2 |
| $(22,4,4)$ | 56 | EXP-007 | pruned | 56 | 27,454 | 0 | 55.7 |
| $(22,4,4)$ | 57 | EXP-007 | pruned | 56 | 190,083 | 0 | 201.8 |
| $(22,4,4)$ | 58 | EXP-007 | pruned | 56 | 1,347,536 | 4 | 1430.5 |
| $(22,4,5)$ | 55 | EXP-007 | pruned | 56 | 2 | 0 | 0.3 |
| $(22,4,5)$ | 56 | EXP-007 | pruned | 56 | 13 | 0 | 16.9 |
| $(22,4,5)$ | 57 | EXP-007 | pruned | 56 | 189 | 0 | 81.3 |
| $(22,4,5)$ | 58 | EXP-007 | pruned | 56 | 2,891 | 0 | 349.7 |
| $(22,4,5)$ | 59 | EXP-007 | pruned | 56 | 44,850 | 0 | 1930.8 |
| $(22,5,3)$ | 38 | EXP-007 | pruned | 56 | 1 | 0 | 5.3 |
| $(22,5,3)$ | 39 | EXP-007 | pruned | 56 | 4 | 0 | 5.1 |
| $(22,5,3)$ | 40 | EXP-007 | pruned | 56 | 22 | 0 | 4.5 |
| $(22,5,3)$ | 41 | EXP-007 | pruned | 56 | 149 | 0 | 4.5 |
| $(22,5,3)$ | 42 | EXP-007 | pruned | 56 | 1,163 | 0 | 8.2 |
| $(22,5,3)$ | 43 | EXP-007 | pruned | 56 | 9,641 | 0 | 12.8 |

| triple | $E$ | record | generator | parts | graphs | kept | seconds |
|---|---|---|---|---|---|---|---|
| $(22,5,3)$ | 44 | EXP-007 | pruned | 56 | 85,229 | 0 | 41.5 |
| $(22,5,3)$ | 45 | EXP-007 | pruned | 56 | 778,285 | 0 | 208.4 |
| $(22,5,3)$ | 46 | EXP-007 | pruned | 56 | 7,249,716 | 31 | 569.7 |
| $(22,5,4)$ | 44 | EXP-007 | pruned | 56 | 3 | 0 | 0.4 |
| $(22,5,4)$ | 45 | EXP-007 | pruned | 56 | 101 | 0 | 31.4 |
| $(22,5,4)$ | 46 | EXP-007 | pruned | 56 | 3,621 | 0 | 262.6 |
| $(22,5,4)$ | 47 | EXP-007 | pruned | 56 | 98,054 | 0 | 1766.8 |
| $(22,6,3)$ | 33 | EXP-007 | pruned | 56 | 1 | 0 | 0.1 |
| $(22,6,3)$ | 34 | EXP-007 | pruned | 56 | 20 | 0 | 0.1 |
| $(22,6,3)$ | 35 | EXP-007 | pruned | 56 | 312 | 0 | 0.6 |
| $(22,6,3)$ | 36 | EXP-007 | pruned | 56 | 5,067 | 0 | 3.3 |
| $(22,6,3)$ | 37 | EXP-007 | pruned | 56 | 80,711 | 0 | 19.6 |
| $(22,6,3)$ | 38 | EXP-007 | pruned | 56 | 1,279,867 | 0 | 89.5 |
| $(22,6,3)$ | 39 | EXP-007 | pruned | 56 | 19,848,521 | 5 | 832.1 |
| $(22,7,3)$ | 33 | EXP-007 | pruned | 56 | 132 | 0 | 0.6 |
| $(22,7,3)$ | 34 | EXP-007 | pruned | 56 | 19,980 | 0 | 799.0 |
| $(23,3,6)$ | 77 | EXP-008 | pruned | 56 | 1 | 0 | 9.4 |
| $(23,3,6)$ | 78 | EXP-008 | pruned | 56 | 3 | 0 | 8.0 |
| $(23,3,6)$ | 79 | EXP-008 | pruned | 56 | 11 | 0 | 8.0 |
| $(23,3,6)$ | 80 | EXP-008 | pruned | 56 | 43 | 0 | 9.6 |
| $(23,3,6)$ | 81 | EXP-008 | pruned | 56 | 184 | 0 | 21.7 |
| $(23,3,6)$ | 82 | EXP-008 | pruned | 56 | 789 | 0 | 41.7 |
| $(23,3,6)$ | 83 | EXP-008 | pruned | 56 | 3,569 | 0 | 87.3 |
| $(23,3,6)$ | 84 | EXP-008 | pruned | 56 | 16,571 | 0 | 270.7 |
| $(23,3,6)$ | 85 | EXP-008 | pruned | 56 | 79,622 | 0 | 849.9 |
| $(23,3,7)$ | 81 | EXP-008 | pruned | 56 | 0 | 0 | 5.8 |
| $(23,3,7)$ | 82 | EXP-008 | pruned | 56 | 0 | 0 | 27.6 |
| $(23,3,7)$ | 83 | EXP-008 | pruned | 56 | 3 | 0 | 70.2 |
| $(23,3,7)$ | 84 | EXP-008 | pruned | 56 | 77 | 0 | 218.8 |
| $(23,3,7)$ | 85 | EXP-008 | pruned | 56 | 756 | 0 | 592.8 |
| $(23,3,7)$ | 86 | EXP-008 | pruned | 56 | 6,742 | 0 | 2579.8 |
| $(23,3,7)$ | 87 | EXP-008 | pruned | 56 | 54,652 | 0 | 10422.5 |
| $(23,4,4)$ | 55 | EXP-008 | pruned | 56 | 1 | 0 | 8.2 |
| $(23,4,4)$ | 56 | EXP-008 | pruned | 56 | 3 | 0 | 10.5 |
| $(23,4,4)$ | 57 | EXP-008 | pruned | 56 | 16 | 0 | 10.8 |
| $(23,4,4)$ | 58 | EXP-008 | pruned | 56 | 92 | 0 | 8.7 |
| $(23,4,4)$ | 59 | EXP-008 | pruned | 56 | 555 | 0 | 12.5 |
| $(23,4,4)$ | 60 | EXP-008 | pruned | 56 | 3,548 | 0 | 19.3 |
| $(23,4,4)$ | 61 | EXP-008 | pruned | 56 | 23,711 | 0 | 57.9 |
| $(23,4,4)$ | 62 | EXP-008 | pruned | 56 | 162,878 | 0 | 214.2 |
| $(23,4,5)$ | 58 | EXP-008 | pruned | 56 | 0 | 0 | 0.9 |
| $(23,4,5)$ | 59 | EXP-008 | pruned | 56 | 2 | 0 | 7.6 |
| $(23,4,5)$ | 60 | EXP-008 | pruned | 56 | 40 | 0 | 21.0 |
| $(23,4,5)$ | 61 | EXP-008 | pruned | 56 | 467 | 0 | 76.7 |
| $(23,4,5)$ | 62 | EXP-008 | pruned | 56 | 5,226 | 0 | 260.1 |
| $(23,4,5)$ | 63 | EXP-008 | pruned | 56 | 54,804 | 0 | 1103.7 |
| $(23,4,5)$ | 64 | EXP-008 | pruned | 56 | 550,018 | 0 | 5880.3 |
| $(23,5,3)$ | 42 | EXP-008 | pruned | 56 | 1 | 0 | 7.2 |
| $(23,5,3)$ | 43 | EXP-008 | pruned | 56 | 4 | 0 | 7.6 |
| $(23,5,3)$ | 44 | EXP-008 | pruned | 56 | 24 | 0 | 7.2 |
| $(23,5,3)$ | 45 | EXP-008 | pruned | 56 | 167 | 0 | 9.2 |
| $(23,5,3)$ | 46 | EXP-008 | pruned | 56 | 1,293 | 0 | 8.5 |
| $(23,5,3)$ | 47 | EXP-008 | pruned | 56 | 10,682 | 0 | 11.7 |
| $(23,5,3)$ | 48 | EXP-008 | pruned | 56 | 93,648 | 0 | 30.5 |
| $(23,5,3)$ | 49 | EXP-008 | pruned | 56 | 848,476 | 0 | 117.8 |
| $(23,5,4)$ | 46 | EXP-008 | pruned | 56 | 2 | 0 | 0.1 |
| $(23,5,4)$ | 47 | EXP-008 | pruned | 56 | 18 | 0 | 3.2 |
| $(23,5,4)$ | 48 | EXP-008 | pruned | 56 | 290 | 0 | 24.1 |
| $(23,5,4)$ | 49 | EXP-008 | pruned | 56 | 4,947 | 0 | 121.1 |
| $(23,5,4)$ | 50 | EXP-008 | pruned | 56 | 83,122 | 0 | 639.5 |
| $(23,5,4)$ | 51 | EXP-008 | pruned | 56 | 1,321,638 | 0 | 4374.0 |
| $(23,6,3)$ | 35 | EXP-008 | pruned | 56 | 1 | 0 | 0.1 |
| $(23,6,3)$ | 36 | EXP-008 | pruned | 56 | 12 | 0 | 0.2 |
| $(23,6,3)$ | 37 | EXP-008 | pruned | 56 | 112 | 0 | 1.1 |
| $(23,6,3)$ | 38 | EXP-008 | pruned | 56 | 1,225 | 0 | 3.2 |
| $(23,6,3)$ | 39 | EXP-008 | pruned | 56 | 14,450 | 0 | 11.3 |

| triple | $E$ | record | generator | parts | graphs | kept | seconds |
|---|---|---|---|---|---|---|---|
| $(23,6,3)$ | 40 | EXP-008 | pruned | 56 | 180,197 | 0 | 36.4 |
| $(23,6,3)$ | 41 | EXP-008 | pruned | 56 | 2,323,048 | 0 | 189.0 |
| $(23,7,3)$ | 35 | EXP-008 | pruned | 56 | 295 | 0 | 17.4 |
| $(23,7,3)$ | 36 | EXP-008 | pruned | 56 | 26,001 | 0 | 603.1 |
| $(23,7,3)$ | 37 | EXP-008 | pruned | 56 | 1,430,451 | 0 | 5905.5 |
| $(24,3,7)$ | 84 | EXP-009 | pruned | 56 | 1 | 0 | 0.9 |
| $(24,3,7)$ | 85 | EXP-009 | pruned | 56 | 1 | 0 | 3.5 |
| $(24,3,7)$ | 86 | EXP-009 | pruned | 56 | 4 | 0 | 11.1 |
| $(24,3,7)$ | 87 | EXP-009 | pruned | 56 | 14 | 0 | 22.2 |
| $(24,3,7)$ | 88 | EXP-009 | pruned | 56 | 56 | 0 | 33.2 |
| $(24,3,7)$ | 89 | EXP-009 | pruned | 56 | 225 | 0 | 51.4 |
| $(24,3,7)$ | 90 | EXP-009 | pruned | 56 | 1,011 | 0 | 133.9 |
| $(24,3,7)$ | 91 | EXP-009 | pruned | 56 | 4,616 | 0 | 372.4 |
| $(24,3,7)$ | 92 | EXP-009 | pruned | 56 | 22,291 | 0 | 1050.2 |
| $(24,3,7)$ | 93 | EXP-009 | pruned | 56 | 110,704 | 0 | 4607.5 |
| $(24,3,7)$ | 94 | EXP-010 | pruned | 56 | 569,264 | 0 | 38875.3 |
| $(24,4,5)$ | 60 | EXP-009 | pruned | 56 | 1 | 0 | 0.1 |
| $(24,4,5)$ | 61 | EXP-009 | pruned | 56 | 1 | 0 | 0.7 |
| $(24,4,5)$ | 62 | EXP-009 | pruned | 56 | 5 | 0 | 4.0 |
| $(24,4,5)$ | 63 | EXP-009 | pruned | 56 | 22 | 0 | 9.8 |
| $(24,4,5)$ | 64 | EXP-009 | pruned | 56 | 116 | 0 | 16.6 |
| $(24,4,5)$ | 65 | EXP-009 | pruned | 56 | 669 | 0 | 25.8 |
| $(24,4,5)$ | 66 | EXP-009 | pruned | 56 | 4,391 | 0 | 58.2 |
| $(24,4,5)$ | 67 | EXP-009 | pruned | 56 | 30,047 | 0 | 237.1 |
| $(24,4,5)$ | 68 | EXP-009 | pruned | 56 | 216,604 | 0 | 1081.0 |
| $(24,4,5)$ | 69 | EXP-009 | pruned | 56 | 1,608,905 | 0 | 6039.7 |
| $(24,5,3)$ | 46 | EXP-009 | pruned | 56 | 1 | 0 | 18.8 |
| $(24,5,3)$ | 47 | EXP-009 | pruned | 56 | 3 | 0 | 18.5 |
| $(24,5,3)$ | 48 | EXP-009 | pruned | 56 | 19 | 0 | 19.1 |
| $(24,5,3)$ | 49 | EXP-009 | pruned | 56 | 125 | 0 | 19.8 |
| $(24,5,3)$ | 50 | EXP-009 | pruned | 56 | 923 | 0 | 20.3 |
| $(24,5,3)$ | 51 | EXP-009 | pruned | 56 | 7,439 | 0 | 23.3 |
| $(24,5,3)$ | 52 | EXP-009 | pruned | 56 | 63,946 | 0 | 50.1 |
| $(24,5,3)$ | 53 | EXP-009 | pruned | 56 | 572,185 | 0 | 199.2 |
| $(24,5,4)$ | 48 | EXP-009 | pruned | 56 | 0 | 0 | 0.1 |
| $(24,5,4)$ | 49 | EXP-009 | pruned | 56 | 2 | 0 | 0.4 |
| $(24,5,4)$ | 50 | EXP-009 | pruned | 56 | 26 | 0 | 2.0 |
| $(24,5,4)$ | 51 | EXP-009 | pruned | 56 | 274 | 0 | 8.9 |
| $(24,5,4)$ | 52 | EXP-009 | pruned | 56 | 3,209 | 0 | 37.3 |
| $(24,5,4)$ | 53 | EXP-009 | pruned | 56 | 38,016 | 0 | 175.7 |
| $(24,5,4)$ | 54 | EXP-009 | pruned | 56 | 453,005 | 0 | 1013.4 |
| $(24,5,4)$ | 55 | EXP-009 | pruned | 56 | 5,355,250 | 0 | 6990.5 |
| $(24,6,3)$ | 36 | EXP-009 | pruned | 56 | 1 | 0 | 0.1 |
| $(24,6,3)$ | 37 | EXP-009 | pruned | 56 | 1 | 0 | 0.2 |
| $(24,6,3)$ | 38 | EXP-009 | pruned | 56 | 5 | 0 | 0.7 |
| $(24,6,3)$ | 39 | EXP-009 | pruned | 56 | 28 | 0 | 3.2 |
| $(24,6,3)$ | 40 | EXP-009 | pruned | 56 | 199 | 0 | 9.9 |
| $(24,6,3)$ | 41 | EXP-009 | pruned | 56 | 1,605 | 0 | 16.0 |
| $(24,6,3)$ | 42 | EXP-009 | pruned | 56 | 15,521 | 0 | 22.7 |
| $(24,6,3)$ | 43 | EXP-009 | pruned | 56 | 165,091 | 0 | 49.5 |
| $(24,6,3)$ | 44 | EXP-009 | pruned | 56 | 1,880,027 | 0 | 210.7 |
| $(24,7,3)$ | 36 | EXP-009 | pruned | 56 | 13 | 0 | 0.2 |
| $(24,7,3)$ | 37 | EXP-009 | pruned | 56 | 556 | 0 | 17.0 |
| $(24,7,3)$ | 38 | EXP-009 | pruned | 56 | 23,292 | 0 | 198.2 |
| $(24,7,3)$ | 39 | EXP-009 | pruned | 56 | 790,718 | 0 | 2066.6 |
| $(24,7,3)$ | 40 | EXP-010 | pruned | 56 | 23,556,793 | 0 | 42833.6 |